

DATABASE MANAGEMENT SYSTEMS

Hospital Management System Project Report

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1. Project Summary

This project is a comprehensive database system designed to streamline the daily operations and record-keeping of a modern hospital. It serves as a central hub for managing detailed information about patients, medical staff, and administrative departments. The system tracks the complete journey of a patient, from maintaining their medical profiles and allergies to scheduling appointments with specialized doctors. It also records medical diagnoses based on international standards and handles the subsequent billing and payment processes. Beyond clinical care, the database organizes staff employment details, doctor specialties, and office locations across various cities and countries. Additionally, it ensures patient safety by linking individuals to emergency contacts and tracking critical health data like blood types. Overall, this integrated platform improves healthcare efficiency by providing a structured and reliable way to manage complex medical and administrative data.

2. Scenario

Department: The Department entity organizes the hospital into functional units. Each department is uniquely identified by a `Department_ID`, which serves as the key attribute. The system also records a unique `department_name`, unique `telephone_number`, a description, and a `head_doctor` (FK). A single department can have multiple doctors assigned to it.

Doctor: The Doctor entity stores professional and personal data for medical staff. Each doctor is uniquely identified by a `Doctor_ID`, which is the key attribute. The system records their `first_name`, `last_name`, `birth_date`, `office_number`, unique `phone_number`, unique email, and gender. It also tracks their `marital_status`, `job_status_id` (FK), `department_id` (FK), and `address_id` (FK). A doctor may also be designated as the head of one department.

Schedule: The Schedule entity manages time slots for hospital availability. Each schedule is uniquely identified by a `Schedule_ID` (PK) and records a specific date and time. A single schedule can be shared by multiple doctors, and the active status of these assignments is stored within the doctor-schedule relationship.

Job Status: The Job Status entity defines the employment condition of staff members. Each job status is uniquely identified by a `Job_Status_ID` (PK) and stores a unique `status_name`. This allows the system to categorize the current working state of every doctor.

Specialization: The Specialization entity tracks various medical fields of expertise. Each specialization is uniquely identified by a `Specialization_ID` and records a unique name. Since doctors may have diverse expertise, a single doctor can hold multiple specializations.

Patient: The Patient entity maintains records for all individuals receiving care. Each patient is uniquely identified by a Patient_ID. The system records their first_name, last_name, birth_date, gender, unique phone_number, and unique email. It also links to a blood_id (FK) and an address_id (FK), and allows each patient to book multiple appointments.

Blood Type: The Blood Type entity categorizes patients based on their biological blood groups. Each blood type has a unique Blood_ID and a unique type_name. This central record allows multiple patients to be associated with the same blood type.

Allergy: The Allergy entity identifies specific medical sensitivities. Each allergy is uniquely identified by an Allergy_ID (PK) and records a unique ICD10_code. Multiple patients can be diagnosed with the same allergy within the system.

Emergency Person: The Emergency Person entity stores contact information for individuals to be reached in case of a patient emergency. Each person is uniquely identified by an Emergency_ID and records their first_name, last_name, and a unique phone_number. An emergency person can be associated with multiple patients.

Address: The Address entity provides detailed geographical locations for staff and patients. Each address is uniquely identified by an Address_ID and includes a country, province, district, neighbourhood, and postal_code. Each specific address record is linked to either one patient or one doctor.

Appointment: The Appointment entity manages the scheduled interactions between patients and staff. Each appointment is uniquely identified by an Appointment_ID and tracks the status, the patient (FK), the doctor (FK), and the schedule (FK). Every appointment record is dedicated to a single patient. The doctor – schedule set is unique.

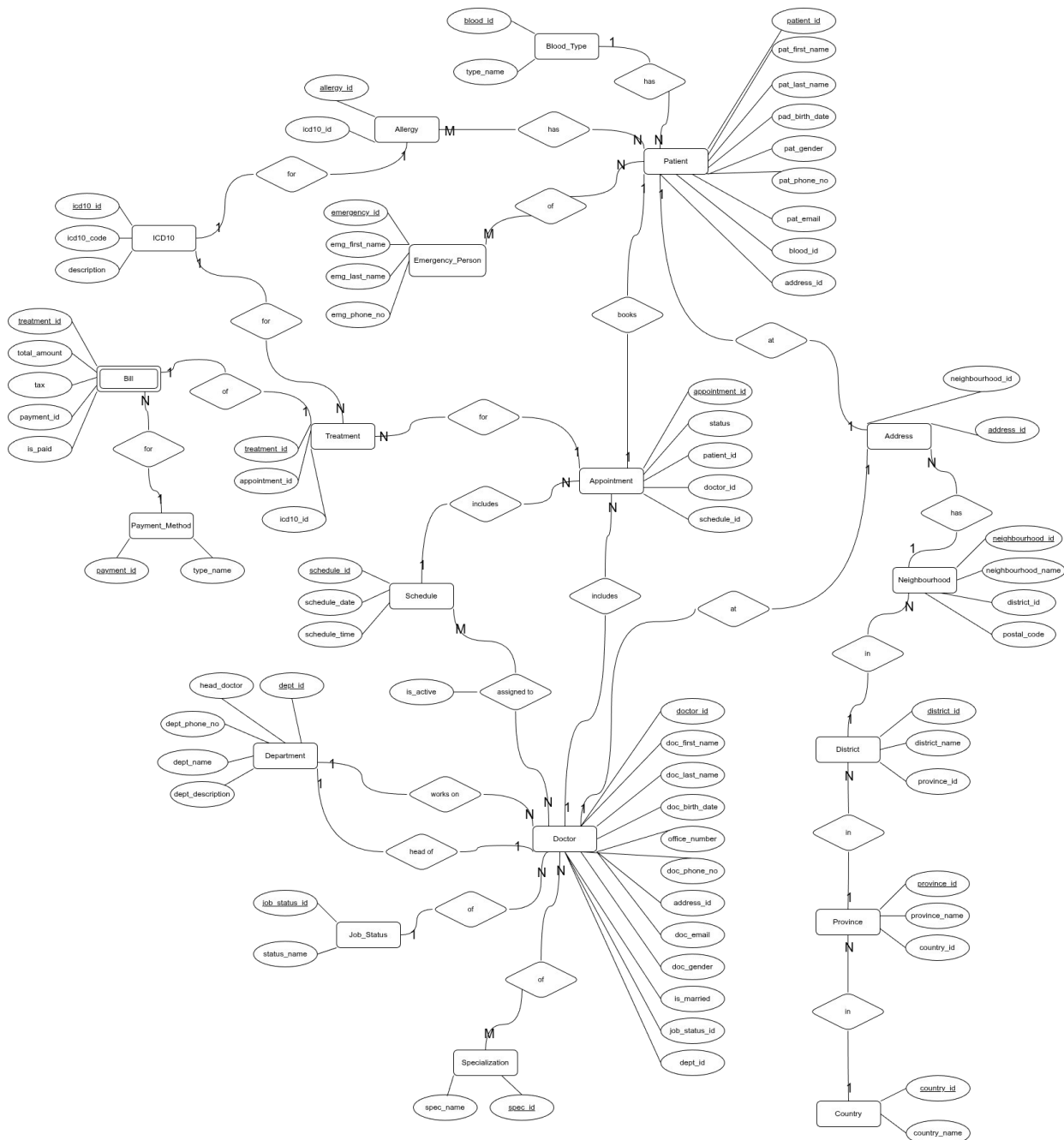
Treatment: The Treatment entity records the specific medical care provided during an appointment. It is identified by a combination of the Appointment_ID and an ICD10_code. Each treatment is strictly tied to one appointment and cannot exist in the system without one.

Bill: The Bill entity handles the financial records for medical treatments. Each bill is uniquely identified by the combination of a Bill_ID and an ICD10_code. The system records the total_amount, tax, payment_id (FK), and whether the bill is_paid. Every bill is directly associated with a specific treatment.

Payment Method: The Payment Method entity defines the accepted ways to settle medical bills. Each method is uniquely identified by a Payment_ID and records a unique type_name. This provides a standardized set of payment options for the billing system.

Relationships: A department can have more than one doctor; a doctor can work in only one department (1:N). A department can have one head doctor; one doctor can be head of one department (1:1). A doctor can have one job status; a job status can be for multiple doctors (1:N). A doctor can have multiple specializations; each specialization can be taken by multiple doctors (M:N). A doctor can have one address; a address can have one doctor (1:1). A doctor can take multiple schedules; each schedule can have multiple doctors (M:N). A doctor can be assigned to multiple appointment; a appointment can be taken by one doctor (1:N). A patient can have one address; a address can have one patient (1:1). A patient can have one blood type; a blood type can be in multiple patients (1:N). A patient can have multiple allergies; a allergy can be in multiple patients (M:N). A patient can have multiple emergency person; each emergency person can have multiple patients (M:N). A appointment can be booked by one patient; a patient can book multiple appointments (1:N). A appointment can include one schedule; a schedule can be for multiple appointments (1:N). A treatment can be created from one appointment; a appointment can create multiple treatments (1:N). A bill can be created from one treatment; a treatment can create one bill (1:1). A bill can have one payment method; each payment method can have multiple bills (1:N).

3. Entity-Relationship Diagram



ERD Description

There are 19 entities and 22 relationships as total. Bill table is weak entity and depends on treatment table because a bill cannot be created without treatment. We have created new tables to handle M:N relationships of tables and named them in the Table1_Table2 format (example: Patient_Allergy).

Since there is a connection table called Doctor_Schedule, which contains doctor_id and schedule_id, in Appointment table we have created a foreign key set (doctor_id, schedule_id) and took those values from Doctor_Schedule table. This way we respect to the triangle relationship (Doctor – Schedule – Appointment) and ensure the uniqueness of doctor – schedule set in the Appointment table.

4. Normalization

Address Table (1NF)

<u>address_id</u>	country	province	district	neighbourhood	postal_code
1	Turkiye	Balikesir	Bandirma	Kayacik	10200
2	England	Manchester	Stockport	Cheadle	SK8 2NP
3	Turkiye	Balikesir	Bandirma	Levent	10200
4	Turkiye	Istanbul	Uskudar	Icadiye	34664

Problems

- Have to insert same columns every time.
- If you update country, you would end up updating all other columns.
- If user with address_id=2 deleted, we will lost England country data.

Functional Dependencies

- address_id → country (partial dependency)
- address_id → province (partial dependency)
- address_id → neighbourhood (partial dependency)
- neighbourhood → district (transitive dependency)
- district → province (transitive dependency)
- province → country (transitive dependency)
- address_id, district → postal_code (key)

Second Normal Form (2NF)

<u>address_id</u>	country	province	neighbourhood
1	Turkiye	Balikesir	Kayacik
2	England	Manchester	Cheadle
3	Turkiye	Balikesir	Levent
4	Turkiye	Istanbul	Icadiye

<u>address_id</u>	<u>district</u>	postal_code
1	Bandirma	10200
2	Stockport	SK8 2NP
3	Bandirma	10200
4	Uskudar	34674

Problems

- We cannot insert values without address_id.
- If we delete address_id, we will delete all data with it.

Third Normal Form (3NF)

Create new tables from transitive dependencies and separate table for country. Add them their own id.

(Table names written below each table.)

<u>country_id</u>	country_name
1	Turkiye
2	England

Country

<u>province_id</u>	province_name	country_id
1	Balikesir	1
2	Manchester	2
3	Istanbul	1

Province

<u>district_id</u>	district_name	province_id
1	Bandirma	1
2	Stockport	2
3	Uskudar	3

District

<u>neighbourhood_id</u>	neighbourhood_name	district_id	postal_code
1	Kayacik	1	10200
2	Cheadle	2	SK8 2NP
3	Levent	1	10200
4	Icadiye	3	34674

Neighbourhood

<u>address_id</u>	neighbourhood_id
1	1
2	2
3	3
4	4

Address

New relationships after normalization: A address can have one neighbourhood; a neighbourhood can be for multiple address (1:N). A neighbourhood can be in one district; a district can have multiple neighbourhoods (1:N). A district can be in one province; a province can have multiple districts (1:N). A province can be in one country; one country can have multiple provinces (1:N).

Treatment Table (3NF)

<u>appointment_id</u>	<u>icd10_code</u>
1	XHF
2	XHF
1	DDA

The table is already in 3NF form, but still there are things we need to consider.

Problems

- If appointment_id is gone, we will lose icd10 code as well.
- You don't know if icd10 code is correct or not. Wrong data insertion problem.

- You may have to write same icd10 code for different appointments.

Functional Dependencies

There is none. You cannot get [appointment → icd10_code] or [icd10_code → appointment_id].

Solution

Create separate table for icd10 codes and give it its own id. Use treatment_id as identifier for Treatment table and take icd10_id and appointment_id as foreign keys. This way Treatment table becomes a form of connection table between Appointment and ICD10 table.

<u>treatment_id</u>	appointment_id	icd10_id
1	1	1
2	2	1
3	1	2

Treatment

<u>icd10_id</u>	icd10_code	description
1	XHF	text1
2	DDA	text2

ICD10

New relationship after normalization: A treatment can have one icd code; a icd code can be for multiple treatments (1:N).

Bill Table (1NF)

<u>bill_id</u>	<u>icd10_code</u>	total_amount	tax	payment_id	is_paid
1	XHF	100	20	null	false
2	XHF	110	30	2	true
3	DDA	200	50	1	true

At first, we have to change icd10_code as treatment_id. Because we have created a separate table for it, and a Bill must be for a Treatment.

Second Normal Form (2NF)

<u>bill_id</u>	treatment_id	total_amount	tax	payment_id	is_paid
1	1	100	20	null	false
2	2	110	30	2	true
3	3	200	50	1	true

Problems

- bill_id and treatment_id are always same. Duplicate column problem.

Solution

Simply delete bill_id and make treatment_id primary key. This way Bill table will become weak and can't be without Treatment table.

Third Normal Form (3NF)

Bill

<u>treatment_id</u>	total_amount	tax	payment_id	is_paid
1	100	20	null	false
2	110	30	2	true
3	200	50	1	true

Relationship remains same.

Allergies (3NF)

<u>allergy_id</u>	icd10_code
1	ABC
2	ABD

Here table is already in 3NF, but we need to change icd10_code as icd10_id.

Final table:

Allergies

<u>allergy_id</u>	icd10_id
1	10
2	11

New relationship after normalization: A allergy can have one icd code; a icd code can define one allergy (1:1).

5. Database Creation

CREATE TABLE

```
CREATE TABLE Country (  
    country_id NUMBER GENERATED BY DEFAULT AS IDENTITY,  
    country_name VARCHAR2(100) NOT NULL,  
    CONSTRAINT pk_country PRIMARY KEY (country_id),  
    CONSTRAINT uq_country_name UNIQUE (country_name)  
);
```

```
CREATE TABLE Province (  
    province_id NUMBER GENERATED BY DEFAULT AS IDENTITY,  
    province_name VARCHAR2(100) NOT NULL,  
    country_id NUMBER NOT NULL,  
    CONSTRAINT pk_province PRIMARY KEY (province_id),  
    CONSTRAINT fk_province_country FOREIGN KEY (country_id) REFERENCES Country(country_id)  
    ON DELETE CASCADE  
);
```

```
CREATE TABLE District (  
    district_id NUMBER GENERATED BY DEFAULT AS IDENTITY,  
    district_name VARCHAR2(100) NOT NULL,  
    province_id NUMBER NOT NULL,
```

```

        CONSTRAINT pk_district PRIMARY KEY (district_id),
        CONSTRAINT fk_district_province FOREIGN KEY (province_id) REFERENCES Province(province_id)
ON DELETE CASCADE
);

```

```

CREATE TABLE Neighbourhood (
    neighbourhood_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    neighbourhood_name VARCHAR2(100) NOT NULL,
    district_id NUMBER NOT NULL,
    postal_code VARCHAR2(20),
    CONSTRAINT pk_neighbourhood PRIMARY KEY (neighbourhood_id),
    CONSTRAINT fk_neighbourhood_district FOREIGN KEY (district_id) REFERENCES
District(district_id) ON DELETE CASCADE
);

```

```

CREATE TABLE Address (
    address_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    neighbourhood_id NUMBER NOT NULL,
    CONSTRAINT pk_address PRIMARY KEY (address_id),
    CONSTRAINT fk_address_neighbourhood FOREIGN KEY (neighbourhood_id) REFERENCES
Neighbourhood(neighbourhood_id) ON DELETE CASCADE
);

```

```

CREATE TABLE Blood_Type (
    blood_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    type_name VARCHAR2(10) NOT NULL,
    CONSTRAINT pk_blood PRIMARY KEY (blood_id),
    CONSTRAINT uq_blood_type_name UNIQUE (type_name),
    CONSTRAINT chk_blood_type_name CHECK(type_name IN ('B+', 'B-', 'A+', 'A-', 'AB+', 'AB-',
'0+', '0-'))
);

```

```

CREATE TABLE ICD10 (
    icd10_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    icd10_code VARCHAR2(20) NOT NULL,
    description CLOB,
    CONSTRAINT pk_icd10 PRIMARY KEY (icd10_id),
    CONSTRAINT uq_icd10_code UNIQUE (icd10_code)
);

```

```

CREATE TABLE Job_Status (
    job_status_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    status_name VARCHAR2(50) NOT NULL,
    CONSTRAINT pk_job_status PRIMARY KEY (job_status_id),
    CONSTRAINT uq_job_status_name UNIQUE (status_name),
    CONSTRAINT chk_job_status_name CHECK(status_name IN ('RETIRED', 'OFFDUTY', 'ACTIVE',
'FIRED'))
);

```

```

CREATE TABLE Specialization (
    spec_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    spec_name VARCHAR2(100) NOT NULL,
    CONSTRAINT pk_specialization PRIMARY KEY (spec_id),
    CONSTRAINT uq_spec_name UNIQUE (spec_name)
);

```

```

CREATE TABLE Payment_Method (
    payment_id NUMBER GENERATED BY DEFAULT AS IDENTITY,

```

```

    type_name VARCHAR2(50) NOT NULL,
    CONSTRAINT pk_payment_method PRIMARY KEY (payment_id),
    CONSTRAINT uq_payment_type_name UNIQUE (type_name)
);

CREATE TABLE Schedule (
    schedule_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    schedule_date DATE NOT NULL,
    schedule_time VARCHAR(5) NOT NULL,
    CONSTRAINT pk_schedule PRIMARY KEY (schedule_id)
);

CREATE TABLE Emergency_Person (
    emergency_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    emg_first_name VARCHAR2(50) NOT NULL,
    emg_last_name VARCHAR2(50) NOT NULL,
    emg_phone_no VARCHAR2(20) NOT NULL,
    emg_gender CHAR(1) NOT NULL,
    CONSTRAINT chk_emg_gender CHECK (emg_gender IN ('M', 'F')),
    CONSTRAINT pk_emergency_person PRIMARY KEY (emergency_id),
    CONSTRAINT uq_emg_phone_no UNIQUE (emg_phone_no)
);

CREATE TABLE Doctor (
    doctor_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    doc_first_name VARCHAR2(50) NOT NULL,
    doc_last_name VARCHAR2(50) NOT NULL,
    doc_birth_date DATE NOT NULL,
    office_number VARCHAR2(20),
    doc_phone_no VARCHAR2(20) NOT NULL,
    doc_email VARCHAR2(100) NOT NULL,
    doc_gender CHAR(1) NOT NULL,
    is_married NUMBER(1), -- 1 for True, 0 for False
    job_status_id NUMBER NOT NULL,
    address_id NUMBER NOT NULL,
    dept_id NUMBER, -- To be handled via ALTER due to circular dependency with Department
    head_doctor
    CONSTRAINT pk_doctor PRIMARY KEY (doctor_id),
    CONSTRAINT uq_doc_phone_no UNIQUE (doc_phone_no),
    CONSTRAINT uq_doc_email UNIQUE (doc_email),
    CONSTRAINT chk_doc_gender CHECK (doc_gender IN ('M', 'F')),
    CONSTRAINT fk_doctor_job_status FOREIGN KEY (job_status_id) REFERENCES
Job_Status(job_status_id),
    CONSTRAINT fk_doctor_address FOREIGN KEY (address_id) REFERENCES Address(address_id)
);

CREATE TABLE Department (
    dept_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    dept_name VARCHAR2(100) NOT NULL,
    dept_phone_no VARCHAR2(20) NOT NULL,
    dept_description CLOB,
    head_doctor NUMBER,
    CONSTRAINT pk_department PRIMARY KEY (dept_id),
    CONSTRAINT uq_dept_name UNIQUE (dept_name),
    CONSTRAINT uq_dept_phone_no UNIQUE (dept_phone_no),
    CONSTRAINT uq_dept_head_doctor UNIQUE (head_doctor),
    CONSTRAINT fk_department_head_doctor FOREIGN KEY (head_doctor) REFERENCES Doctor(doctor_id)
ON DELETE SET NULL

```

);

```
ALTER TABLE Doctor ADD CONSTRAINT fk_doctor_department FOREIGN KEY (dept_id) REFERENCES
Department(dept_id) ON DELETE SET NULL;
```

```
CREATE TABLE Patient (
    patient_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    pat_first_name VARCHAR2(50) NOT NULL,
    pat_last_name VARCHAR2(50) NOT NULL,
    pat_birth_date DATE NOT NULL,
    pat_gender CHAR(1) NOT NULL,
    pat_phone_no VARCHAR2(20) NOT NULL,
    pat_email VARCHAR2(100) NOT NULL,
    blood_id NUMBER NOT NULL,
    address_id NUMBER NOT NULL,
    CONSTRAINT pk_patient PRIMARY KEY (patient_id),
    CONSTRAINT uq_pat_phone_no UNIQUE (pat_phone_no),
    CONSTRAINT uq_pat_email UNIQUE (pat_email),
    CONSTRAINT chk_pat_gender CHECK (pat_gender IN ('M', 'F')),
    CONSTRAINT fk_patient_blood FOREIGN KEY (blood_id) REFERENCES Blood_Type(blood_id),
    CONSTRAINT fk_patient_address FOREIGN KEY (address_id) REFERENCES Address(address_id)
);
```

```
CREATE TABLE Doctor_Schedule (
    doctor_id NUMBER NOT NULL,
    schedule_id NUMBER NOT NULL,
    is_active NUMBER(1) DEFAULT 1 NOT NULL,
    CONSTRAINT pk_doctor_schedule PRIMARY KEY (doctor_id, schedule_id),
    CONSTRAINT fk_doc_sch_doctor FOREIGN KEY (doctor_id) REFERENCES Doctor(doctor_id) ON DELETE
CASCADE,
    CONSTRAINT fk_doc_sch_schedule FOREIGN KEY (schedule_id) REFERENCES Schedule(schedule_id)
ON DELETE CASCADE
);
```

```
CREATE TABLE Appointment (
    appointment_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    is_active NUMBER(1) DEFAULT 1 NOT NULL,
    patient_id NUMBER NOT NULL,
    doctor_id NUMBER NOT NULL,
    schedule_id NUMBER NOT NULL,
    CONSTRAINT pk_appointment PRIMARY KEY (appointment_id),
    CONSTRAINT fk_appointment_patient FOREIGN KEY (patient_id) REFERENCES Patient(patient_id)
ON DELETE CASCADE,
    CONSTRAINT fk_appointment_doc_sch FOREIGN KEY (doctor_id, schedule_id) REFERENCES
Doctor_Schedule(doctor_id, schedule_id) ON DELETE CASCADE,
    CONSTRAINT uq_appointment_doc_sch UNIQUE (doctor_id, schedule_id)
);
```

```
CREATE TABLE Treatment (
    treatment_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    appointment_id NUMBER NOT NULL,
    icd10_id NUMBER NOT NULL,
    CONSTRAINT pk_treatment PRIMARY KEY (treatment_id),
    CONSTRAINT fk_treatment_appointment FOREIGN KEY (appointment_id) REFERENCES
Appointment(appointment_id) ON DELETE CASCADE,
    CONSTRAINT fk_treatment_icd10 FOREIGN KEY (icd10_id) REFERENCES ICD10(icd10_id) ON DELETE
CASCADE
);
```

```

CREATE TABLE Bill (
    treatment_id NUMBER,
    total_amount NUMBER(10, 2) NOT NULL,
    tax NUMBER(10, 2) NOT NULL,
    payment_id NUMBER,
    is_paid NUMBER(1) DEFAULT 0 NOT NULL,
    CONSTRAINT pk_bill PRIMARY KEY (treatment_id),
    CONSTRAINT chk_bill_total_amount CHECK (total_amount >= 0),
    CONSTRAINT chk_bill_tax CHECK (tax >= 0),
    CONSTRAINT fk_bill_treatment FOREIGN KEY (treatment_id) REFERENCES Treatment(treatment_id)
ON DELETE CASCADE,
    CONSTRAINT fk_bill_payment_method FOREIGN KEY (payment_id) REFERENCES
Payment_Method(payment_id) ON DELETE SET NULL
);

```

```

CREATE TABLE Allergy (
    allergy_id NUMBER GENERATED BY DEFAULT AS IDENTITY,
    icd10_id NUMBER NOT NULL,
    CONSTRAINT pk_allergy PRIMARY KEY (allergy_id),
    CONSTRAINT uq_allergy_icd10 UNIQUE (icd10_id),
    CONSTRAINT fk_allergy_icd10 FOREIGN KEY (icd10_id) REFERENCES ICD10(icd10_id) ON DELETE
CASCADE
);

```

```

CREATE TABLE Doctor_Specialization (
    doctor_id NUMBER NOT NULL,
    spec_id NUMBER NOT NULL,
    CONSTRAINT pk_doctor_specialization PRIMARY KEY (doctor_id, spec_id),
    CONSTRAINT fk_doc_spec_doctor FOREIGN KEY (doctor_id) REFERENCES Doctor(doctor_id) ON
DELETE CASCADE,
    CONSTRAINT fk_doc_spec_specialization FOREIGN KEY (spec_id) REFERENCES
Specialization(spec_id) ON DELETE CASCADE
);

```

```

CREATE TABLE Patient_Allergy (
    patient_id NUMBER NOT NULL,
    allergy_id NUMBER NOT NULL,
    CONSTRAINT pk_patient_allergy PRIMARY KEY (patient_id, allergy_id),
    CONSTRAINT fk_pat_allergy_patient FOREIGN KEY (patient_id) REFERENCES Patient(patient_id)
ON DELETE CASCADE,
    CONSTRAINT fk_pat_allergy_allergy FOREIGN KEY (allergy_id) REFERENCES Allergy(allergy_id)
ON DELETE CASCADE
);

```

```

CREATE TABLE Patient_Emergency_Person (
    patient_id NUMBER NOT NULL,
    emergency_id NUMBER NOT NULL,
    CONSTRAINT pk_patient_emergency_person PRIMARY KEY (patient_id, emergency_id),
    CONSTRAINT fk_pat_emg_patient FOREIGN KEY (patient_id) REFERENCES Patient(patient_id) ON
DELETE CASCADE,
    CONSTRAINT fk_pat_emg_emergency FOREIGN KEY (emergency_id) REFERENCES
Emergency_Person(emergency_id) ON DELETE CASCADE
);

```

```
Script Output x
Task completed in 1.219 seconds

Table COUNTRY created.

Table PROVINCE created.

Table DISTRICT created.

Table NEIGHBOURHOOD created.

Table ADDRESS created.

Table BLOOD_TYPE created.

Table ICD10 created.

Table JOB_STATUS created.

Table SPECIALIZATION created.

Table PAYMENT_METHOD created.

Table SCHEDULE created.

Table EMERGENCY_PERSON created.
```

```
Script Output x
Task completed in 1.219 seconds

Table DOCTOR created.

Table DEPARTMENT created.

Table DOCTOR altered.

Table PATIENT created.

Table DOCTOR_SCHEDULE created.

Table APPOINTMENT created.

Table TREATMENT created.

Table BILL created.

Table ALLERGY created.

Table DOCTOR_SPECIALIZATION created.

Table PATIENT_ALLERGY created.

Table PATIENT_EMERGENCY_PERSON created.
```

INSERT INTO — Sample Data

```
INSERT INTO Country (country_name) VALUES ('Turkiye');
INSERT INTO Country (country_name) VALUES ('USA');
INSERT INTO Country (country_name) VALUES ('UK'
);
INSERT INTO Country (country_name) VALUES ('Germany');
INSERT INTO Country (country_name) VALUES ('France');
INSERT INTO Country (country_name) VALUES ('Italy');
INSERT INTO Country (country_name) VALUES ('Spain');
INSERT INTO Country (country_name) VALUES ('Japan');
INSERT INTO Country (country_name) VALUES ('China');
INSERT INTO Country (country_name) VALUES ('Canada');
INSERT INTO Country (country_name) VALUES ('Australia');
INSERT INTO Country (country_name) VALUES ('Brazil');
INSERT INTO Country (country_name) VALUES ('Russia');
INSERT INTO Country (country_name) VALUES ('India');
INSERT INTO Country (country_name) VALUES ('Netherlands');
INSERT INTO Country (country_name) VALUES ('Switzerland');
INSERT INTO Country (country_name) VALUES ('Sweden');
INSERT INTO Country (country_name) VALUES ('Norway');
INSERT INTO Country (country_name) VALUES ('Denmark');
INSERT INTO Country (country_name) VALUES ('Belgium');
INSERT INTO Country (country_name) VALUES ('Austria');
INSERT INTO Country (country_name) VALUES ('Portugal');
INSERT INTO Country (country_name) VALUES ('Greece');
INSERT INTO Country (country_name) VALUES ('South Korea');
INSERT INTO Country (country_name) VALUES ('Mexico');
```

```
INSERT INTO Province (province_name, country_id) VALUES ('Istanbul', 1);
INSERT INTO Province (province_name, country_id) VALUES ('Ankara', 1);
INSERT INTO Province (province_name, country_id) VALUES ('Izmir', 1);
INSERT INTO Province (province_name, country_id) VALUES ('New York', 2);
INSERT INTO Province (province_name, country_id) VALUES ('California', 2);
INSERT INTO Province (province_name, country_id) VALUES ('Texas', 2);
INSERT INTO Province (province_name, country_id) VALUES ('London', 3);
INSERT INTO Province (province_name, country_id) VALUES ('Manchester', 3);
INSERT INTO Province (province_name, country_id) VALUES ('Berlin', 4);
INSERT INTO Province (province_name, country_id) VALUES ('Munich', 4);
INSERT INTO Province (province_name, country_id) VALUES ('Paris', 5);
```

```

INSERT INTO Province (province_name, country_id) VALUES ('Rome', 6);
INSERT INTO Province (province_name, country_id) VALUES ('Madrid', 7);
INSERT INTO Province (province_name, country_id) VALUES ('Tokyo', 8);
INSERT INTO Province (province_name, country_id) VALUES ('Beijing', 9);
INSERT INTO Province (province_name, country_id) VALUES ('Toronto', 10);
INSERT INTO Province (province_name, country_id) VALUES ('Sydney', 11);
INSERT INTO Province (province_name, country_id) VALUES ('Sao Paulo', 12);
INSERT INTO Province (province_name, country_id) VALUES ('Moscow', 13);
INSERT INTO Province (province_name, country_id) VALUES ('Mumbai', 14);
INSERT INTO Province (province_name, country_id) VALUES ('Amsterdam', 15);
INSERT INTO Province (province_name, country_id) VALUES ('Zurich', 16);
INSERT INTO Province (province_name, country_id) VALUES ('Stockholm', 17);
INSERT INTO Province (province_name, country_id) VALUES ('Oslo', 18);
INSERT INTO Province (province_name, country_id) VALUES ('Copenhagen', 19);

INSERT INTO District (district_name, province_id) VALUES ('Kadikoy', 1);
INSERT INTO District (district_name, province_id) VALUES ('Besiktas', 1);
INSERT INTO District (district_name, province_id) VALUES ('Cankaya', 2);
INSERT INTO District (district_name, province_id) VALUES ('Konak', 3);
INSERT INTO District (district_name, province_id) VALUES ('Manhattan', 4);
INSERT INTO District (district_name, province_id) VALUES ('Brooklyn', 4);
INSERT INTO District (district_name, province_id) VALUES ('Los Angeles', 5);
INSERT INTO District (district_name, province_id) VALUES ('Houston', 6);
INSERT INTO District (district_name, province_id) VALUES ('Westminster', 7);
INSERT INTO District (district_name, province_id) VALUES ('Camden', 7);
INSERT INTO District (district_name, province_id) VALUES ('Salford', 8);
INSERT INTO District (district_name, province_id) VALUES ('Mitte', 9);
INSERT INTO District (district_name, province_id) VALUES ('Charlottenburg', 9);
INSERT INTO District (district_name, province_id) VALUES ('Maxvorstadt', 10);
INSERT INTO District (district_name, province_id) VALUES ('Le Marais', 11);
INSERT INTO District (district_name, province_id) VALUES ('Trastevere', 12);
INSERT INTO District (district_name, province_id) VALUES ('Retiro', 13);
INSERT INTO District (district_name, province_id) VALUES ('Shinjuku', 14);
INSERT INTO District (district_name, province_id) VALUES ('Chaoyang', 15);
INSERT INTO District (district_name, province_id) VALUES ('North York', 16);
INSERT INTO District (district_name, province_id) VALUES ('CBD', 17);
INSERT INTO District (district_name, province_id) VALUES ('Pinheiros', 18);
INSERT INTO District (district_name, province_id) VALUES ('Arbat', 19);
INSERT INTO District (district_name, province_id) VALUES ('Bandra West', 20);
INSERT INTO District (district_name, province_id) VALUES ('Centrum', 21);

INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Moda', 1,
'34710');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Caferaga', 1,
'34711');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Etiler', 2,
'34337');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES
('Bahcelievler', 3, '06490');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Alsancak', 4,
'35220');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Upper East
Side', 5, '10021');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES
('Williamsburg', 6, '11211');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Santa
Monica', 7, '90401');

```

```

INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Downtown', 8, '77002');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Soho', 9, 'W1D');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Covent Garden', 9, 'WC2H');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Primrose Hill', 10, 'NW1');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Quays', 11, 'M50');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Alexanderplatz', 12, '10178');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Tiergarten', 12, '10557');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Savignyplatz', 13, '10623');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Univ-Quarter', 14, '80799');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Saint-Paul', 15, '75004');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Janiculum', 16, '00153');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Jerónimos', 17, '28014');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Kabukicho', 18, '160-0021');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Sanlitun', 19, '100027');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Willowdale', 20, 'M2N');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('The Rocks', 21, '2000');
INSERT INTO Neighbourhood (neighbourhood_name, district_id, postal_code) VALUES ('Vila Madalena', 22, '05445-000');

```

```

INSERT INTO Address (neighbourhood_id)
SELECT TRUNC(DBMS_RANDOM.VALUE(1, 26)) FROM DUAL CONNECT BY LEVEL <= 150;

```

```

INSERT INTO Blood_Type (type_name) VALUES ('A+');
INSERT INTO Blood_Type (type_name) VALUES ('A-');
INSERT INTO Blood_Type (type_name) VALUES ('B+');
INSERT INTO Blood_Type (type_name) VALUES ('B-');
INSERT INTO Blood_Type (type_name) VALUES ('AB+');
INSERT INTO Blood_Type (type_name) VALUES ('AB-');
INSERT INTO Blood_Type (type_name) VALUES ('O+');
INSERT INTO Blood_Type (type_name) VALUES ('O-');

```

```

INSERT INTO ICD10 (icd10_code, description) VALUES ('I10', 'Essential (primary) hypertension');
INSERT INTO ICD10 (icd10_code, description) VALUES ('E11.9', 'Type 2 diabetes mellitus without complications');
INSERT INTO ICD10 (icd10_code, description) VALUES ('J06.9', 'Acute upper respiratory infection, unspecified');
INSERT INTO ICD10 (icd10_code, description) VALUES ('M54.5', 'Low back pain');
INSERT INTO ICD10 (icd10_code, description) VALUES ('K21.9', 'Gastro-esophageal reflux disease without esophagitis');
INSERT INTO ICD10 (icd10_code, description) VALUES ('F32.9', 'Major depressive disorder, single episode, unspecified');

```

```

INSERT INTO ICD10 (icd10_code, description) VALUES ('N39.0', 'Urinary tract infection, site not
specified');
INSERT INTO ICD10 (icd10_code, description) VALUES ('G43.909', 'Migraine, unspecified, not
intractable, without status migsignosus');
INSERT INTO ICD10 (icd10_code, description) VALUES ('Z00.00', 'Encounter for general adult
medical examination without abnormal findings');
INSERT INTO ICD10 (icd10_code, description) VALUES ('L20.9', 'Atopic dermatitis, unspecified');
INSERT INTO ICD10 (icd10_code, description) VALUES ('H10.9', 'Unspecified conjunctivitis');
INSERT INTO ICD10 (icd10_code, description) VALUES ('I25.10', 'Atherosclerotic heart disease of
native coronary artery without angina pectoris');
INSERT INTO ICD10 (icd10_code, description) VALUES ('M17.9', 'Osteoarthritis of knee,
unspecified');
INSERT INTO ICD10 (icd10_code, description) VALUES ('R05', 'Cough');
INSERT INTO ICD10 (icd10_code, description) VALUES ('F41.1', 'Generalized anxiety disorder');
INSERT INTO ICD10 (icd10_code, description) VALUES ('E03.9', 'Hypothyroidism, unspecified');
INSERT INTO ICD10 (icd10_code, description) VALUES ('B35.1', 'Tinea unguium');
INSERT INTO ICD10 (icd10_code, description) VALUES ('N40.0', 'Benign prostatic hyperplasia
without lower urinary tract symptoms');
INSERT INTO ICD10 (icd10_code, description) VALUES ('M54.2', 'Cervicalgia');
INSERT INTO ICD10 (icd10_code, description) VALUES ('R51', 'Headache');
INSERT INTO ICD10 (icd10_code, description) VALUES ('J45.909', 'Unspecified asthma,
uncomplicated');
INSERT INTO ICD10 (icd10_code, description) VALUES ('K29.70', 'Gastritis, unspecified, without
bleeding');
INSERT INTO ICD10 (icd10_code, description) VALUES ('M51.26', 'Other intervertebral disc
displacement, lumbar region');
INSERT INTO ICD10 (icd10_code, description) VALUES ('Q67.4', 'Other congenital deformities of
skull, face and jaw');
INSERT INTO ICD10 (icd10_code, description) VALUES ('S06.0X0A', 'Concussion without loss of
consciousness, initial encounter');

```

```

INSERT INTO Job_Status (status_name) VALUES ('ACTIVE');
INSERT INTO Job_Status (status_name) VALUES ('RETIRED');
INSERT INTO Job_Status (status_name) VALUES ('OFFDUTY');
INSERT INTO Job_Status (status_name) VALUES ('FIRED');

```

```

INSERT INTO Specialization (spec_name) VALUES ('Cardiology');
INSERT INTO Specialization (spec_name) VALUES ('Internal Medicine');
INSERT INTO Specialization (spec_name) VALUES ('Pediatrics');
INSERT INTO Specialization (spec_name) VALUES ('Neurology');
INSERT INTO Specialization (spec_name) VALUES ('Dermatology');
INSERT INTO Specialization (spec_name) VALUES ('Psychiatry');
INSERT INTO Specialization (spec_name) VALUES ('Radiology');
INSERT INTO Specialization (spec_name) VALUES ('Orthopedics');
INSERT INTO Specialization (spec_name) VALUES ('Ophthalmology');
INSERT INTO Specialization (spec_name) VALUES ('General Surgery');
INSERT INTO Specialization (spec_name) VALUES ('Urology');
INSERT INTO Specialization (spec_name) VALUES ('Gynecology');
INSERT INTO Specialization (spec_name) VALUES ('Endocrinology');
INSERT INTO Specialization (spec_name) VALUES ('Oncology');
INSERT INTO Specialization (spec_name) VALUES ('Nephrology');
INSERT INTO Specialization (spec_name) VALUES ('Pulmonology');
INSERT INTO Specialization (spec_name) VALUES ('Gastroenterology');
INSERT INTO Specialization (spec_name) VALUES ('Rheumatology');
INSERT INTO Specialization (spec_name) VALUES ('Otolaryngology');
INSERT INTO Specialization (spec_name) VALUES ('Anesthesiology');
INSERT INTO Specialization (spec_name) VALUES ('Pathology');
INSERT INTO Specialization (spec_name) VALUES ('Emergency Medicine');

```

```

INSERT INTO Specialization (spec_name) VALUES ('Physical Medicine');
INSERT INTO Specialization (spec_name) VALUES ('Medical Genetics');
INSERT INTO Specialization (spec_name) VALUES ('Infectious Disease');

```

```

INSERT INTO Payment_Method (type_name) VALUES ('Cash');
INSERT INTO Payment_Method (type_name) VALUES ('Credit Card');
INSERT INTO Payment_Method (type_name) VALUES ('Insurance');
INSERT INTO Payment_Method (type_name) VALUES ('Bank Transfer');
INSERT INTO Payment_Method (type_name) VALUES ('Apple Pay');
INSERT INTO Payment_Method (type_name) VALUES ('Google Pay');
INSERT INTO Payment_Method (type_name) VALUES ('PayPal');
INSERT INTO Payment_Method (type_name) VALUES ('Venmo');
INSERT INTO Payment_Method (type_name) VALUES ('Zelle');
INSERT INTO Payment_Method (type_name) VALUES ('Cashier's Check');
INSERT INTO Payment_Method (type_name) VALUES ('Personal Check');
INSERT INTO Payment_Method (type_name) VALUES ('Bitcoin');
INSERT INTO Payment_Method (type_name) VALUES ('Ethereum');
INSERT INTO Payment_Method (type_name) VALUES ('Litecoin');
INSERT INTO Payment_Method (type_name) VALUES ('Klarna');
INSERT INTO Payment_Method (type_name) VALUES ('Affirm');
INSERT INTO Payment_Method (type_name) VALUES ('Afterpay');
INSERT INTO Payment_Method (type_name) VALUES ('Wise');
INSERT INTO Payment_Method (type_name) VALUES ('Revolut');
INSERT INTO Payment_Method (type_name) VALUES ('Cash App');
INSERT INTO Payment_Method (type_name) VALUES ('Skrill');
INSERT INTO Payment_Method (type_name) VALUES ('Neteller');
INSERT INTO Payment_Method (type_name) VALUES ('Western Union');
INSERT INTO Payment_Method (type_name) VALUES ('MoneyGram');
INSERT INTO Payment_Method (type_name) VALUES ('Payoneer');

```

```

INSERT INTO Schedule (schedule_date, schedule_time)
WITH hours AS (SELECT level + 7 AS h FROM dual CONNECT BY level <= 10),
     mins AS (SELECT (level - 1) * 15 AS m FROM dual CONNECT BY level <= 4)
SELECT DATE '2026-06-03',
       LPAD(h, 2, '0') || ':' || LPAD(m, 2, '0')
FROM hours CROSS JOIN mins
WHERE h <> 12;

```

```

INSERT INTO Schedule (schedule_date, schedule_time)
WITH hours AS (SELECT level + 7 AS h FROM dual CONNECT BY level <= 10),
     mins AS (SELECT (level - 1) * 15 AS m FROM dual CONNECT BY level <= 4)
SELECT DATE '2026-06-08',
       LPAD(h, 2, '0') || ':' || LPAD(m, 2, '0')
FROM hours CROSS JOIN mins
WHERE h <> 12;

```

```

INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('John', 'Doe', '555-0001', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Jane', 'Smith', '555-0002', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Mehmet', 'Yilmaz', '555-0003', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Ayse', 'Kaya', '555-0004', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Robert', 'Brown', '555-0005', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Emily', 'Davis', '555-0006', 'F');

```

```

INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Michael', 'Wilson', '555-0007', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Sarah', 'Miller', '555-0008', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('David', 'Taylor', '555-0009', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Linda', 'Anderson', '555-0010', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('James', 'Moore', '555-0011', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Patricia', 'Taylor', '555-0012', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Christopher', 'Thomas', '555-0013', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Barbara', 'Jackson', '555-0014', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Thomas', 'Anderson', '555-0015', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Sarah', 'Connor', '555-0016', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Bruce', 'Wayne', '555-0017', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Clark', 'Kent', '555-0018', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Diana', 'Prince', '555-0019', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Peter', 'Parker', '555-0020', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Tony', 'Stark', '555-0021', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Natasha', 'Romanoff', '555-0022', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Steve', 'Rogers', '555-0023', 'M');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Wanda', 'Maximoff', '555-0024', 'F');
INSERT INTO Emergency_Person (emg_first_name, emg_last_name, emg_phone_no, emg_gender) VALUES
('Bucky', 'Barnes', '555-0025', 'M');

```

```

INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Gregory',
'House', DATE '1959-05-15', '101', '555-1001', 'house@hospital.com', 'M', 0, 1, 1, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('James',
'Wilson', DATE '1967-02-28', '102', '555-1002', 'wilson@hospital.com', 'M', 0, 1, 2, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Lisa', 'Cuddy',
DATE '1966-07-03', '103', '555-1003', 'cuddy@hospital.com', 'F', 0, 1, 3, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Eric',
'Foreman', DATE '1973-07-23', '104', '555-1004', 'foreman@hospital.com', 'M', 0, 1, 4, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Allison',
'Cameron', DATE '1979-04-12', '105', '555-1005', 'cameron@hospital.com', 'F', 1, 1, 5, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Robert',
'Chase', DATE '1979-02-12', '106', '555-1006', 'chase@hospital.com', 'M', 0, 1, 6, NULL);

```

```

INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Chris', 'Taub',
DATE '1967-10-10', '107', '555-1007', 'taub@hospital.com', 'M', 1, 1, 7, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Remy',
'Hadley', DATE '1984-03-10', '108', '555-1008', 'thirteen@hospital.com', 'F', 0, 1, 8, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Lawrence',
'Kutner', DATE '1975-01-01', '109', '555-1009', 'kutner@hospital.com', 'M', 0, 1, 9, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Martha',
'Masters', DATE '1988-11-11', '110', '555-1010', 'masters@hospital.com', 'F', 0, 1, 10, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Chi', 'Park',
DATE '1985-12-12', '111', '555-1011', 'park@hospital.com', 'F', 0, 1, 11, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Jessica',
'Adams', DATE '1982-06-06', '112', '555-1012', 'adams@hospital.com', 'F', 0, 1, 12, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Amber',
'Volakis', DATE '1980-01-20', '113', '555-1013', 'amber@hospital.com', 'F', 0, 1, 13, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Lucas',
'Douglas', DATE '1975-03-15', '114', '555-1014', 'lucas@hospital.com', 'M', 0, 1, 14, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Stacy',
'Warner', DATE '1965-09-30', '115', '555-1015', 'stacy@hospital.com', 'F', 1, 1, 15, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Stephen',
'Strange', DATE '1976-11-18', '201', '555-1016', 'strange@hospital.com', 'M', 0, 1, 16, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('John',
'Watson', DATE '1971-08-16', '202', '555-1017', 'watson@hospital.com', 'M', 1, 1, 17, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Dana',
'Scully', DATE '1964-02-23', '203', '555-1018', 'scully@hospital.com', 'F', 0, 1, 18, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Leonard',
'McCoy', DATE '1920-01-20', '204', '555-1019', 'bones@hospital.com', 'M', 0, 1, 19, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Beverly',
'Crusher', DATE '1948-10-13', '205', '555-1020', 'crusher@hospital.com', 'F', 1, 1, 20, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Juliet',
'Douglas', DATE '1980-05-05', '206', '555-1021', 'juliet@hospital.com', 'F', 0, 1, 21, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Jack',
'Shephard', DATE '1969-07-20', '207', '555-1022', 'shephard@hospital.com', 'M', 0, 1, 22,
NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Christian',
'Shephard', DATE '1945-12-15', '208', '555-1023', 'christian@hospital.com', 'M', 1, 1, 23,
NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Meredith',
'Grey', DATE '1978-11-10', '209', '555-1024', 'grey@hospital.com', 'F', 1, 1, 24, NULL);
INSERT INTO Doctor (doc_first_name, doc_last_name, doc_birth_date, office_number, doc_phone_no,
doc_email, doc_gender, is_married, job_status_id, address_id, dept_id) VALUES ('Derek',

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'Shepherd', DATE '1966-08-16', '210', '555-1025', 'mcdreamy@hospital.com', 'M', 1, 1, 25,
NULL);
```

```
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Diagnostic Medicine', '555-2001', 'Specializes in difficult cases.', 1);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Oncology', '555-2002', 'Cancer treatment and research.', 2);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Administration', '555-2003', 'Hospital management.', 3);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Neurology', '555-2004', 'Nervous system disorders.', 4);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Immunology', '555-2005', 'Immune system studies.', 5);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Surgery', '555-2006', 'General surgical procedures.', 6);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Emergency', '555-2007', 'Acute care services.', 12);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Cardiology', '555-2008', 'Heart health and treatment.', 16);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Pediatrics', '555-2009', 'Child healthcare.', 17);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Dermatology', '555-2010', 'Skin conditions.', 18);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Psychiatry', '555-2011', 'Mental health services.', 19);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Radiology', '555-2012', 'Imaging and diagnostics.', 20);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Orthopedics', '555-2013', 'Bone and joint care.', 21);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Ophthalmology', '555-2014', 'Eye care.', 22);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('General Surgery', '555-2015', 'Surgical interventions.', 23);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Urology', '555-2016', 'Urinary tract health.', 24);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Gynecology', '555-2017', 'Women's health.', 25);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Endocrinology', '555-2018', 'Hormone disorders.', 11);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Nephrology', '555-2019', 'Kidney health.', 8);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Pulmonology', '555-2020', 'Respiratory system.', 9);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Gastroenterology', '555-2021', 'Digestive system.', 10);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Rheumatology', '555-2022', 'Joint and muscle diseases.', 13);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Otolaryngology', '555-2023', 'Ear, nose, and throat.', 14);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Anesthesiology', '555-2024', 'Pain management and anesthesia.', 15);
INSERT INTO Department (dept_name, dept_phone_no, dept_description, head_doctor) VALUES
('Pathology', '555-2025', 'Laboratory diagnostics.', 7);
```

```
UPDATE Doctor SET dept_id = 1 WHERE doctor_id IN (1, 8, 9, 10, 11);
UPDATE Doctor SET dept_id = 2 WHERE doctor_id = 2;
UPDATE Doctor SET dept_id = 3 WHERE doctor_id = 3;
UPDATE Doctor SET dept_id = 4 WHERE doctor_id = 4;
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UPDATE Doctor SET dept_id = 5 WHERE doctor_id = 5;
UPDATE Doctor SET dept_id = 6 WHERE doctor_id IN (6, 7);
UPDATE Doctor SET dept_id = 7 WHERE doctor_id IN (12, 13, 14, 15);
UPDATE Doctor SET dept_id = 8 WHERE doctor_id = 16;
UPDATE Doctor SET dept_id = 9 WHERE doctor_id = 17;
UPDATE Doctor SET dept_id = 10 WHERE doctor_id = 18;
UPDATE Doctor SET dept_id = 11 WHERE doctor_id = 19;
UPDATE Doctor SET dept_id = 12 WHERE doctor_id = 20;
UPDATE Doctor SET dept_id = 13 WHERE doctor_id = 21;
UPDATE Doctor SET dept_id = 14 WHERE doctor_id = 22;
UPDATE Doctor SET dept_id = 15 WHERE doctor_id = 23;
UPDATE Doctor SET dept_id = 16 WHERE doctor_id = 24;
UPDATE Doctor SET dept_id = 17 WHERE doctor_id = 25;

```

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INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Alice', 'Wonderland', DATE '1995-03-12', 'F', '555-
3001', 'alice@wonder.com', 1, 16);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Bob', 'Builder', DATE '1980-11-20', 'M', '555-3002',
'bob@build.com', 3, 17);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Charlie', 'Brown', DATE '2010-10-30', 'M', '555-
3003', 'charlie@peanuts.com', 7, 18);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Diana', 'Prince', DATE '1985-05-20', 'F', '555-3004',
'diana@themyscira.com', 5, 19);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Edward', 'Norton', DATE '1969-08-18', 'M', '555-
3005', 'edward@fightclub.com', 2, 20);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Fiona', 'Shrek', DATE '1990-04-05', 'F', '555-3006',
'fiona@swamp.com', 4, 21);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('George', 'Costanza', DATE '1959-12-15', 'M', '555-
3007', 'george@vandelay.com', 8, 22);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Hannah', 'Baker', DATE '2000-01-01', 'F', '555-3008',
'hannah@tapes.com', 6, 23);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Ian', 'McKellen', DATE '1939-05-25', 'M', '555-3009',
'ian@gandalf.com', 1, 24);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Julia', 'Roberts', DATE '1967-10-28', 'F', '555-
3010', 'julia@hollywood.com', 3, 25);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Kevin', 'Spacey', DATE '1959-07-26', 'M', '555-3011',
'kevin@usualsuspects.com', 5, 26);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Lara', 'Croft', DATE '1992-02-14', 'F', '555-3012',
'lara@tombraider.com', 2, 27);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Mickey', 'Mouse', DATE '1928-11-18', 'M', '555-3013',
'mickey@disney.com', 4, 28);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Nancy', 'Drew', DATE '2005-05-05', 'F', '555-3014',
'nancy@mystery.com', 8, 29);

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INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Oscar', 'Grouch', DATE '1969-11-10', 'M', '555-3015',
'oscar@sesame.com', 6, 30);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Peter', 'Parker', DATE '2001-08-10', 'M', '555-3016',
'peter@dailybugle.com', 1, 31);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Quinn', 'Harley', DATE '1990-07-20', 'F', '555-3017',
'quinn@arkham.com', 3, 32);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Riley', 'Reid', DATE '1991-07-09', 'F', '555-3018',
'riley@test.com', 7, 33);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Steve', 'Rogers', DATE '1918-07-04', 'M', '555-3019',
'steve@avengers.com', 5, 34);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Tony', 'Stark', DATE '1970-05-29', 'M', '555-3020',
'tony@starkindustries.com', 2, 35);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Ursula', 'Sea', DATE '1975-02-02', 'F', '555-3021',
'ursula@disney.com', 4, 36);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Victor', 'Frankenstein', DATE '1800-01-01', 'M',
'555-3022', 'victor@monster.com', 8, 37);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Wanda', 'Maximoff', DATE '1989-02-10', 'F', '555-
3023', 'wanda@avengers.com', 6, 38);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Xavier', 'Charles', DATE '1940-05-05', 'M', '555-
3024', 'xavier@xmen.com', 1, 39);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Yoda', 'Grand', DATE '0900-01-01', 'M', '555-3025',
'yoda@jedi.com', 3, 40);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Zelda', 'Hyrule', DATE '1998-02-21', 'F', '555-3026',
'zelda@nintendo.com', 5, 41);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Arthur', 'Morgan', DATE '1863-01-01', 'M', '555-
3027', 'arthur@rdr2.com', 2, 42);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Bella', 'Swan', DATE '1987-09-13', 'F', '555-3028',
'bella@twilight.com', 4, 43);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Clark', 'Kent', DATE '1978-06-18', 'M', '555-3029',
'clark@dailyplanet.com', 8, 44);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Don', 'Draper', DATE '1926-06-01', 'M', '555-3030',
'don@madmen.com', 6, 45);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Ellen', 'Ripley', DATE '2092-01-07', 'F', '555-3031',
'ripley@alien.com', 1, 46);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Frodo', 'Baggins', DATE '2968-09-22', 'M', '555-
3032', 'frodo@shire.com', 3, 47);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Geralt', 'Rivia', DATE '1200-01-01', 'M', '555-3033',
'geralt@witcher.com', 5, 48);

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INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Harry', 'Potter', DATE '1980-07-31', 'M', '555-3034',
'harry@hogwarts.com', 2, 49);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Iris', 'West', DATE '1990-06-24', 'F', '555-3035',
'iris@centralcity.com', 4, 50);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Joel', 'Miller', DATE '1981-09-26', 'M', '555-3036',
'joel@lastofus.com', 8, 51);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Katniss', 'Everdeen', DATE '1994-05-08', 'F', '555-
3037', 'katniss@district12.com', 6, 52);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Luke', 'Skywalker', DATE '1951-09-25', 'M', '555-
3038', 'luke@tatooine.com', 1, 53);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Mary', 'Jane', DATE '1982-08-15', 'F', '555-3039',
'mj@dailybugle.com', 3, 54);
INSERT INTO Patient (pat_first_name, pat_last_name, pat_birth_date, pat_gender, pat_phone_no,
pat_email, blood_id, address_id) VALUES ('Neo', 'Anderson', DATE '1962-03-11', 'M', '555-3040',
'neo@matrix.com', 5, 55);

```

```

INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (1, 1, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (1, 2, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (1, 3, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (1, 4, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (2, 5, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (2, 6, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (2, 7, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (2, 8, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (3, 9, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (3, 10, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (4, 11, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (4, 12, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (5, 13, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (5, 14, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (6, 15, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (6, 16, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (7, 17, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (7, 18, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (8, 19, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (8, 20, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (9, 21, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (10, 22, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (11, 23, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (12, 24, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (13, 25, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (14, 26, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (15, 27, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (1, 28, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (2, 29, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (3, 30, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (4, 31, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (5, 32, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (6, 33, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (7, 34, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (8, 35, 1);
INSERT INTO Doctor_Schedule (doctor_id, schedule_id, is_active) VALUES (9, 36, 1);

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[illegible]

```
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 1, 1, 1);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 2, 1, 2);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 3, 1, 3);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 4, 1, 4);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 5, 2, 5);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 6, 2, 6);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 7, 2, 7);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 8, 2, 8);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 9, 3, 9);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 10, 3, 10);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 11, 4, 11);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 12, 4, 12);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 13, 5, 13);
```



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INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 31, 14, 67);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 32, 15, 68);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 33, 1, 69);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 34, 2, 70);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 35, 3, 71);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 36, 4, 72);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 37, 5, 37);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 38, 6, 38);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 39, 7, 39);
INSERT INTO Appointment (is_active, patient_id, doctor_id, schedule_id) VALUES (1, 40, 8, 40);

```

```

INSERT INTO Treatment (appointment_id, icd10_id)
SELECT appointment_id, TRUNC(DBMS_RANDOM.VALUE(1, 26)) FROM Appointment;

```

```

INSERT INTO Bill (treatment_id, total_amount, tax, payment_id, is_paid)
SELECT treatment_id,
       100 + DBMS_RANDOM.VALUE(0, 900),
       20 + DBMS_RANDOM.VALUE(0, 180),
       TRUNC(DBMS_RANDOM.VALUE(1, 26)),
       CASE WHEN DBMS_RANDOM.VALUE(0, 1) > 0.4 THEN 1 ELSE 0 END
FROM Treatment;

```

```

INSERT INTO Allergy (icd10_id) VALUES (1);
INSERT INTO Allergy (icd10_id) VALUES (2);
INSERT INTO Allergy (icd10_id) VALUES (3);
INSERT INTO Allergy (icd10_id) VALUES (4);
INSERT INTO Allergy (icd10_id) VALUES (5);
INSERT INTO Allergy (icd10_id) VALUES (6);
INSERT INTO Allergy (icd10_id) VALUES (7);
INSERT INTO Allergy (icd10_id) VALUES (8);
INSERT INTO Allergy (icd10_id) VALUES (9);
INSERT INTO Allergy (icd10_id) VALUES (10);
INSERT INTO Allergy (icd10_id) VALUES (11);
INSERT INTO Allergy (icd10_id) VALUES (12);
INSERT INTO Allergy (icd10_id) VALUES (13);
INSERT INTO Allergy (icd10_id) VALUES (14);
INSERT INTO Allergy (icd10_id) VALUES (15);
INSERT INTO Allergy (icd10_id) VALUES (16);
INSERT INTO Allergy (icd10_id) VALUES (17);
INSERT INTO Allergy (icd10_id) VALUES (18);
INSERT INTO Allergy (icd10_id) VALUES (19);
INSERT INTO Allergy (icd10_id) VALUES (20);
INSERT INTO Allergy (icd10_id) VALUES (21);
INSERT INTO Allergy (icd10_id) VALUES (22);
INSERT INTO Allergy (icd10_id) VALUES (23);
INSERT INTO Allergy (icd10_id) VALUES (24);
INSERT INTO Allergy (icd10_id) VALUES (25);

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```

INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (1, 1);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (1, 2);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (2, 2);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (3, 3);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (4, 4);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (5, 5);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (6, 6);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (7, 7);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (8, 8);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (9, 9);

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INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (10, 10);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (11, 4);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (12, 10);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (13, 10);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (14, 11);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (15, 12);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (16, 14);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (17, 3);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (18, 5);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (19, 6);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (20, 7);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (21, 8);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (22, 9);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (23, 10);
INSERT INTO Doctor_Specialization (doctor_id, spec_id) VALUES (24, 11);

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INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (1, 1);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (3, 2);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (8, 3);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (12, 4);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (15, 1);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (22, 5);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (30, 2);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (35, 4);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (38, 3);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (40, 5);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (2, 6);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (4, 7);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (5, 8);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (6, 9);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (7, 11);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (9, 12);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (10, 13);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (11, 14);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (13, 16);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (14, 17);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (16, 18);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (17, 19);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (18, 21);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (19, 22);
INSERT INTO Patient_Allergy (patient_id, allergy_id) VALUES (20, 23);

```

```

INSERT INTO Patient_Emergency_Person (patient_id, emergency_id)
SELECT p.patient_id, TRUNC(DBMS_RANDOM.VALUE(1, 26)) FROM Patient p;

```

```

COMMIT;

```

Country

COUNTRY_ID	COUNTRY_NAME
1	Turkiye
2	USA
3	UK
4	Germany
5	France
6	Italy
7	Spain
8	Japan
9	China
10	Canada
11	Australia
12	Brazil
13	Russia
14	India
15	Netherlands

Province

PROVINCE_ID	PROVINCE_NAME	COUNTRY_ID
1	Istanbul	1
2	Ankara	1
3	Izmir	1
4	New York	2
5	California	2
6	Texas	2
7	London	3
8	Manchester	3
9	Berlin	4
10	Munich	4
11	Paris	5
12	Rome	6
13	Madrid	7
14	Tokyo	8
15	Beijing	9

District

DISTRICT_ID	DISTRICT_NAME	PROVINCE_ID
1	Kadikoy	1
2	Besiktas	1
3	Cankaya	2
4	Konak	3
5	Manhattan	4
6	Brooklyn	4
7	Los Angeles	5
8	Houston	6
9	Westminster	7
10	Camden	7
11	Salford	8
12	Mitte	9
13	Charlottenburg	9
14	Maxvorstadt	10
15	Le Marais	11

Neighbourhood

NEIGHBOURHOOD_ID	NEIGHBOURHOOD_NAME	DISTRICT_ID	POSTAL_CODE
1	Moda	1	34710
2	Caferaga	1	34711
3	Etiler	2	34337
4	Bahcelievler	3	06490
5	Alsancak	4	35220
6	Upper East Side	5	10021
7	Williamsburg	6	11211
8	Santa Monica	7	90401
9	Downtown	8	77002
10	Soho	9	W1D
11	Covent Garden	9	WC2H
12	Primrose Hill	10	NW1
13	Quays	11	M50
14	Alexanderplatz	12	10178
15	Tiergarten	12	10557

Address

ADDRESS_ID	NEIGHBOURHOOD_ID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15

Blood_Type

BLOOD_ID	TYPE_NAME
1	A+
2	A-
3	B+
4	B-
5	AB+
6	AB-
7	O+
8	O-

ICD10

ICD10_ID	ICD10_CODE	DESCRIPTION
1	I10	Essential (primary) hypertension
2	E11.9	Type 2 diabetes mellitus without complications
3	J06.9	Acute upper respiratory infection, unspecified
4	M54.5	Low back pain
5	K21.9	Gastro-esophageal reflux disease without esophagitis
6	F32.9	Major depressive disorder, single episode, unspecified
7	N39.0	Urinary tract infection, site not specified
8	G43.909	Migraine, unspecified, not intractable, without status migsignosus
9	Z00.00	Encounter for general adult medical examination without abnormal findings
10	L20.9	Atopic dermatitis, unspecified
11	H10.9	Unspecified conjunctivitis
12	I25.10	Atherosclerotic heart disease of native coronary artery without angina p...
13	M17.9	Osteoarthritis of knee, unspecified
14	R05	Cough
15	F41.1	Generalized anxiety disorder

Job_Status

JOB_STATUS_ID	STATUS_NAME
1	ACTIVE
2	RETIRED
3	OFFDUTY
4	FIRE

Specialization

SPEC_ID	SPEC_NAME
1	1 Cardiology
2	2 Internal Medicine
3	3 Pediatrics
4	4 Neurology
5	5 Dermatology
6	6 Psychiatry
7	7 Radiology
8	8 Orthopedics
9	9 Ophthalmology
10	10 General Surgery
11	11 Urology
12	12 Gynecology
13	13 Endocrinology
14	14 Oncology
15	15 Nephrology

Payment_Method

PAYMENT_ID	TYPE_NAME
1	1 Cash
2	2 Credit Card
3	3 Insurance
4	4 Bank Transfer
5	5 Apple Pay
6	6 Google Pay
7	7 PayPal
8	8 Venmo
9	9 Zelle
10	10 Cashier's Check
11	11 Personal Check
12	12 Bitcoin
13	13 Ethereum
14	14 Litecoin
15	15 Klarna
16	16 Affirm
17	17 Afterpay
18	18 Wise
19	19 Revolut
20	20 Cash App
21	21 Skrill
22	22 Neteller
23	23 Western Union

Schedule

SCHEDULE_ID	SCHEDULE_DATE	SCHEDULE_TIME
1	1 03-JUN-26	08:00
2	2 03-JUN-26	08:15
3	3 03-JUN-26	08:30
4	4 03-JUN-26	08:45
5	5 03-JUN-26	09:00
6	6 03-JUN-26	09:15
7	7 03-JUN-26	09:30
8	8 03-JUN-26	09:45
9	9 03-JUN-26	10:00
10	10 03-JUN-26	10:15
11	11 03-JUN-26	10:30
12	12 03-JUN-26	10:45
13	13 03-JUN-26	11:00
14	14 03-JUN-26	11:15
15	15 03-JUN-26	11:30
16	16 03-JUN-26	11:45
17	17 03-JUN-26	13:00
18	18 03-JUN-26	13:15
19	19 03-JUN-26	13:30
20	20 03-JUN-26	13:45

Allergy

ALLERGY_ID	ICD 10_ID
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23

Emergency_Person

EMERGENCY_ID	EMG_FIRST_NAME	EMG_LAST_NAME	EMG_PHONE_NO	EMG_GENDER
1	1 John	Doe	555-0001	M
2	2 Jane	Smith	555-0002	F
3	3 Mehmet	Yilmaz	555-0003	M
4	4 Ayse	Kaya	555-0004	F
5	5 Robert	Brown	555-0005	M
6	6 Emily	Davis	555-0006	F
7	7 Michael	Wilson	555-0007	M
8	8 Sarah	Miller	555-0008	F
9	9 David	Taylor	555-0009	M
10	10 Linda	Anderson	555-0010	F
11	11 James	Moore	555-0011	M
12	12 Patricia	Taylor	555-0012	F
13	13 Christopher	Thomas	555-0013	M
14	14 Barbara	Jackson	555-0014	F
15	15 Thomas	Anderson	555-0015	M
16	16 Sarah	Connor	555-0016	F
17	17 Bruce	Wayne	555-0017	M
18	18 Clark	Kent	555-0018	M
19	19 Diana	Prince	555-0019	F
20	20 Peter	Parker	555-0020	M

Doctor

DOCTOR_ID	DOC_FIRST_NAME	DOC_LAST_NAME	DOC_BIRTH_DATE	OFFICE_NUMBER	DOC_PHONE_NO	DOC_EMAIL	DOC_GENDER	IS_MARRIED	JOB_STATUS_ID	ADDRESS_ID	DEPT_ID
1	Gregory	House	15-MAY-59	101	555-1001	house@hospital.com	M	0	1	1	1
2	James	Wilson	28-FEB-67	102	555-1002	wilson@hospital.com	M	0	1	2	2
3	Lisa	Cuddy	03-JUL-66	103	555-1003	cuddy@hospital.com	F	0	1	3	3
4	Eric	Foreman	23-JUL-73	104	555-1004	foreman@hospital.com	M	0	1	4	4
5	Allison	Cameron	12-APR-79	105	555-1005	cameron@hospital.com	F	1	1	5	5
6	Robert	Chase	12-FEB-79	106	555-1006	chase@hospital.com	M	0	1	6	6
7	Chris	Taub	10-OCT-67	107	555-1007	taub@hospital.com	M	1	1	7	6
8	Remy	Hadley	10-MAR-84	108	555-1008	thirteen@hospital.com	F	0	1	8	1
9	Lawrence	Kutner	01-JAN-75	109	555-1009	kutner@hospital.com	M	0	1	9	1
10	Martha	Masters	11-NOV-88	110	555-1010	masters@hospital.com	F	0	1	10	1
11	Chi	Park	12-DEC-85	111	555-1011	park@hospital.com	F	0	1	11	1
12	Jessica	Adams	06-JUN-82	112	555-1012	adams@hospital.com	F	0	1	12	7
13	Amber	Volakis	20-JAN-80	113	555-1013	amber@hospital.com	F	0	1	13	7
14	Lucas	Douglas	15-MAR-75	114	555-1014	lucas@hospital.com	M	0	1	14	7
15	Stacy	Warner	30-SEP-65	115	555-1015	stacy@hospital.com	F	1	1	15	7
16	Stephen	Strange	18-NOV-76	201	555-1016	strange@hospital.com	M	0	1	16	8
17	John	Watson	16-AUG-71	202	555-1017	watson@hospital.com	M	1	1	17	9
18	Dana	Scully	23-FEB-64	203	555-1018	scully@hospital.com	F	0	1	18	10
19	Leonard	McCoy	20-JAN-20	204	555-1019	bones@hospital.com	M	0	1	19	11
20	Beverly	Crusher	13-OCT-48	205	555-1020	crusher@hospital.com	F	1	1	20	12
21	Juliet	Douglas	05-MAY-80	206	555-1021	juliet@hospital.com	F	0	1	21	13
22	Jack	Shephard	20-JUL-69	207	555-1022	shephard@hospital.com	M	0	1	22	14
23	Christian	Shephard	15-DEC-45	208	555-1023	christian@hospital.com	M	1	1	23	15
24	Meredith	Grey	10-NOV-78	209	555-1024	grey@hospital.com	F	1	1	24	16
25	Derek	Shepherd	16-AUG-66	210	555-1025	mcdreamy@hospital.com	M	1	1	25	17

Department

DEPT_ID	DEPT_NAME	DEPT_PHONE_NO	DEPT_DESCRIPTION	HEAD_DOCTOR
1	1 Diagnostic Medicine	555-2001	Specializes in difficult cases.	1
2	2 Oncology	555-2002	Cancer treatment and research.	2
3	3 Administration	555-2003	Hospital management.	3
4	4 Neurology	555-2004	Nervous system disorders.	4
5	5 Immunology	555-2005	Immune system studies.	5
6	6 Surgery	555-2006	General surgical procedures.	6
7	7 Emergency	555-2007	Acute care services.	12
8	8 Cardiology	555-2008	Heart health and treatment.	16
9	9 Pediatrics	555-2009	Child healthcare.	17
10	10 Dermatology	555-2010	Skin conditions.	18
11	11 Psychiatry	555-2011	Mental health services.	19
12	12 Radiology	555-2012	Imaging and diagnostics.	20
13	13 Orthopedics	555-2013	Bone and joint care.	21
14	14 Ophthalmology	555-2014	Eye care.	22
15	15 General Surgery	555-2015	Surgical interventions.	23
16	16 Urology	555-2016	Urinary tract health.	24
17	17 Gynecology	555-2017	Women's health.	25
18	18 Endocrinology	555-2018	Hormone disorders.	11
19	19 Nephrology	555-2019	Kidney health.	8
20	20 Pulmonology	555-2020	Respiratory system.	9
21	21 Gastroenterology	555-2021	Digestive system.	10
22	22 Rheumatology	555-2022	Joint and muscle diseases.	13
23	23 Otolaryngology	555-2023	Ear, nose, and throat.	14
24	24 Anesthesiology	555-2024	Pain management and anesthesia.	15
25	25 Pathology	555-2025	Laboratory diagnostics.	7

Patient

PATIENT_ID	PAT_FIRST_NAME	PAT_LAST_NAME	PAT_BIRTH_DATE	PAT_GENDER	PAT_PHONE_NO	PAT_EMAIL	BLOOD_ID	ADDRESS_ID
1	Alice	Wonderland	12-MAR-95	F	555-3001	alice@wonder.com	1	16
2	Bob	Builder	20-NOV-80	M	555-3002	bob@build.com	3	17
3	Charlie	Brown	30-OCT-10	M	555-3003	charlie@peanuts.com	7	18
4	Diana	Prince	20-MAY-85	F	555-3004	diana@themyscira.com	5	19
5	Edward	Norton	18-AUG-69	M	555-3005	edward@fightclub.com	2	20
6	Fiona	Shrek	05-APR-90	F	555-3006	fiona@swamp.com	4	21
7	George	Costanza	15-DEC-59	M	555-3007	george@vandelay.com	8	22
8	Hannah	Baker	01-JAN-00	F	555-3008	hannah@tapes.com	6	23
9	Ian	McKellen	25-MAY-39	M	555-3009	ian@gandalf.com	1	24
10	Julia	Roberts	28-OCT-67	F	555-3010	julia@hollywood.com	3	25
11	Kevin	Spacey	26-JUL-59	M	555-3011	kevin@usualsuspects.com	5	26
12	Lara	Croft	14-FEB-92	F	555-3012	lara@tombrainer.com	2	27
13	Mickey	Mouse	18-NOV-28	M	555-3013	mickey@disney.com	4	28
14	Nancy	Drew	05-MAY-05	F	555-3014	nancy@mystery.com	8	29
15	Oscar	Grouch	10-NOV-69	M	555-3015	oscar@sesame.com	6	30
16	Peter	Parker	10-AUG-01	M	555-3016	peter@dailybugle.com	1	31
17	Quinn	Harley	20-JUL-90	F	555-3017	quinn@arkham.com	3	32
18	Riley	Reid	09-JUL-91	F	555-3018	riley@test.com	7	33
19	Steve	Rogers	04-JUL-18	M	555-3019	steve@avengers.com	5	34
20	Tony	Stark	29-MAY-70	M	555-3020	tony@starkindustries.com	2	35
21	Ursula	Sea	02-FEB-75	F	555-3021	ursula@disney.com	4	36
22	Victor	Frankenstein	01-JAN-00	M	555-3022	victor@monster.com	8	37
23	Wanda	Maximoff	10-FEB-89	F	555-3023	wanda@avengers.com	6	38
24	Xavier	Charles	05-MAY-40	M	555-3024	xavier@xmen.com	1	39
25	Yoda	Grand	01-JAN-00	M	555-3025	yoda@iedi.com	3	40

Doctor_Schedule

DOCTOR_ID	SCHEDULE_ID	IS_ACTIVE
1	1	1
2	1	2
3	1	3
4	1	4
5	2	5
6	2	6
7	2	7
8	2	8
9	3	9
10	3	10
11	4	11
12	4	12
13	5	13
14	5	14
15	6	15
16	6	16
17	7	17
18	7	18
19	8	19
20	8	20
21	9	21
22	10	22
23	11	23
24	12	24
25	13	25

Appointment

APPOINTMENT_ID	IS_ACTIVE	PATIENT_ID	DOCTOR_ID	SCHEDULE_ID
1	1	1	1	1
2	2	2	1	2
3	3	3	1	3
4	4	4	1	4
5	5	5	2	5
6	6	6	2	6
7	7	7	2	7
8	8	8	2	8
9	9	9	3	9
10	10	10	3	10
11	11	11	4	11
12	12	12	4	12
13	13	13	5	13
14	14	14	5	14
15	15	15	6	15
16	16	16	6	16
17	17	17	7	17
18	18	18	7	18
19	19	19	8	19
20	20	20	8	20
21	21	21	9	21
22	22	22	10	22
23	23	23	11	23
24	24	24	12	24
25	25	25	13	25

Treatment

TREATMENT_ID	APPOINTMENT_ID	ICD10_ID
1	1	5
2	2	1
3	3	18
4	4	9
5	5	11
6	6	19
7	7	10
8	8	5
9	9	21
10	10	14
11	11	25
12	12	3
13	13	5
14	14	8
15	15	21
16	16	9
17	17	13
18	18	21
19	19	10
20	20	14
21	21	18
22	22	15
23	23	2
24	24	19
25	25	11

Bill

TREATMENT_ID	TOTAL_AMOUNT	TAX	PAYMENT_ID	IS_PAID
1	937.92	140.71	15	0
2	669.86	137.53	23	1
3	757.04	111.02	1	1
4	429.75	108.7	6	1
5	737.88	23.89	19	1
6	365.67	84.59	19	0
7	295.61	153.39	9	1
8	460.59	76.74	11	0
9	389.05	68.62	3	0
10	323.44	96.89	17	0
11	352.72	26.74	24	1
12	369.03	35.33	9	1
13	453.8	157.38	17	0
14	902.05	129.31	25	0
15	470.49	185.59	14	0
16	971.33	83.2	1	0
17	798.18	39.11	20	1
18	927.96	180.57	2	1
19	153.17	141.15	24	1
20	499.61	32.5	15	1
21	553.84	23.05	22	0
22	930.24	91.9	5	1
23	271.02	133.45	5	0
24	800.44	145.88	11	0
25	229.25	146.89	2	0

Patient_Allergy

PATIENT_ID	ALLERGY_ID
1	1
2	3
3	8
4	12
5	15
6	22
7	30
8	35
9	38
10	40
11	2
12	4
13	5
14	6
15	7
16	9
17	10
18	11
19	13
20	14
21	16
22	17
23	18
24	19
25	20

Patient_Emergency_Person

PATIENT_ID	EMERGENCY_ID
1	8
2	18
3	21
4	22
5	12
6	3
7	22
8	14
9	6
10	1
11	23
12	25
13	1
14	13
15	11
16	6
17	20
18	14
19	11
20	23
21	5
22	1
23	19
24	10
25	23

Doctor_Specialization

DOCTOR_ID	SPEC_ID
1	1
2	2
3	2
4	3
5	4
6	5
7	6
8	7
9	8
10	9
11	10
12	4
13	10
14	10
15	11
16	12
17	14
18	3
19	5
20	6

6. SQL Queries

Yılmaz

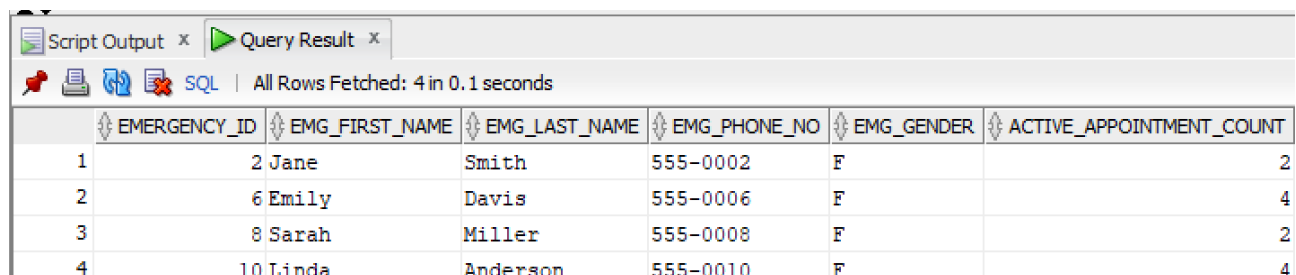
Problem

List emergency persons and their patient's active appointment counts where the count is bigger than zero and patients have blood type "A+".

Query

```
SELECT ep.*,
       COUNT(a.APPOINTMENT_ID) AS active_appointment_count
FROM EMERGENCY_PERSON ep
JOIN PATIENT_EMERGENCY_PERSON pep ON ep.EMERGENCY_ID = pep.EMERGENCY_ID
JOIN APPOINTMENT a ON pep.PATIENT_ID = a.PATIENT_ID
WHERE a.IS_ACTIVE = 1
      AND ep.EMERGENCY_ID IN (
        SELECT sub_pep.EMERGENCY_ID
        FROM PATIENT_EMERGENCY_PERSON sub_pep
        JOIN PATIENT p ON sub_pep.PATIENT_ID = p.PATIENT_ID
        WHERE p.BLOOD_ID = (
          SELECT blood_id FROM BLOOD_TYPE
          WHERE type_name = 'A+'
        )
      )
GROUP BY ep.EMERGENCY_ID, ep.EMG_FIRST_NAME, ep.EMG_LAST_NAME,
ep.EMG_PHONE_NO, ep.EMG_GENDER
HAVING COUNT(a.APPOINTMENT_ID) > 0
ORDER BY ep.EMERGENCY_ID ASC;
```

Results



The screenshot shows a SQL query result window with a tab labeled 'Query Result'. Below the tab, it says 'All Rows Fetched: 4 in 0.1 seconds'. The table has 6 columns: EMERGENCY_ID, EMG_FIRST_NAME, EMG_LAST_NAME, EMG_PHONE_NO, EMG_GENDER, and ACTIVE_APPOINTMENT_COUNT. The data is as follows:

	EMERGENCY_ID	EMG_FIRST_NAME	EMG_LAST_NAME	EMG_PHONE_NO	EMG_GENDER	ACTIVE_APPOINTMENT_COUNT
1	2	Jane	Smith	555-0002	F	2
2	6	Emily	Davis	555-0006	F	4
3	8	Sarah	Miller	555-0008	F	2
4	10	Linda	Anderson	555-0010	F	4

Problem

List patients and their financial impact analysis based on treatment costs where patients have registered allergies and their average spending exceeds the hospital-wide average.

Query

```
SELECT p.PAT_FIRST_NAME, p.PAT_LAST_NAME, p.PAT_EMAIL,
       COUNT(t.TREATMENT_ID) AS treatment_count,
       SUM(b.TOTAL_AMOUNT) AS total_spending
FROM PATIENT p
```

```

JOIN APPOINTMENT a ON p.PATIENT_ID = a.PATIENT_ID
JOIN TREATMENT t ON a.APPOINTMENT_ID = t.APPOINTMENT_ID
JOIN BILL b ON t.TREATMENT_ID = b.TREATMENT_ID
WHERE b.IS_PAID = 1
      AND p.PATIENT_ID IN (
          SELECT sub_pa.PATIENT_ID
          FROM PATIENT_ALLERGY sub_pa
      )
GROUP BY p.PATIENT_ID, p.PAT_FIRST_NAME, p.PAT_LAST_NAME, p.PAT_EMAIL
HAVING AVG(b.TOTAL_AMOUNT) > (
    SELECT AVG(total_amount) FROM BILL
)
ORDER BY total_spending DESC;

```

Results

Script Output x

Query Result x

   SQL | All Rows Fetched: 9 in 0.003 seconds

	PAT_FIRST_NAME	PAT_LAST_NAME	PAT_EMAIL	TREATMENT_COUNT	TOTAL_SPENDING
1	Riley	Reid	riley@test.com	2	1872.17
2	Neo	Anderson	neo@matrix.com	2	1692.93
3	Iris	West	iris@centralcity.com	2	1409.61
4	Bob	Builder	bob@build.com	2	1303.31
5	Edward	Norton	edward@fightclub.com	2	1278.86
6	Victor	Frankenstein	victor@monster.com	1	930.24
7	Luke	Skywalker	luke@tatooine.com	1	924.84
8	Mickey	Mouse	mickey@disney.com	1	832.59
9	Ian	McKellen	ian@gandalf.com	1	655.51

Heba

Problem

List count of active doctors that are located in each district where count is bigger than zero.

Query

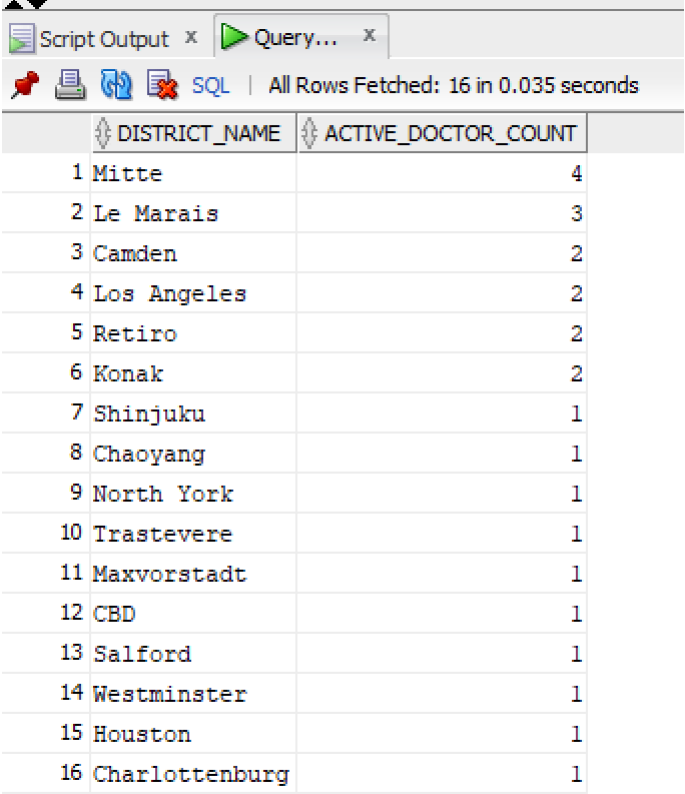
```

SELECT d.district_name,
       COUNT(doc.doctor_id) AS active_doctor_count
FROM District d
JOIN Neighbourhood n ON d.district_id = n.district_id
JOIN Address a ON n.neighbourhood_id = a.neighbourhood_id
JOIN Doctor doc ON a.address_id = doc.address_id
WHERE doc.job_status_id = (
    SELECT job_status_id
    FROM Job_Status
    WHERE status_name = 'ACTIVE'
)
GROUP BY d.district_name

```

```
HAVING COUNT(doc.doctor_id) > 0
ORDER BY active_doctor_count DESC;
```

Results



DISTRICT_NAME	ACTIVE_DOCTOR_COUNT
1 Mitte	4
2 Le Marais	3
3 Camden	2
4 Los Angeles	2
5 Retiro	2
6 Konak	2
7 Shinjuku	1
8 Chaoyang	1
9 North York	1
10 Trastevere	1
11 Maxvorstadt	1
12 CBD	1
13 Salford	1
14 Westminster	1
15 Houston	1
16 Charlottenburg	1

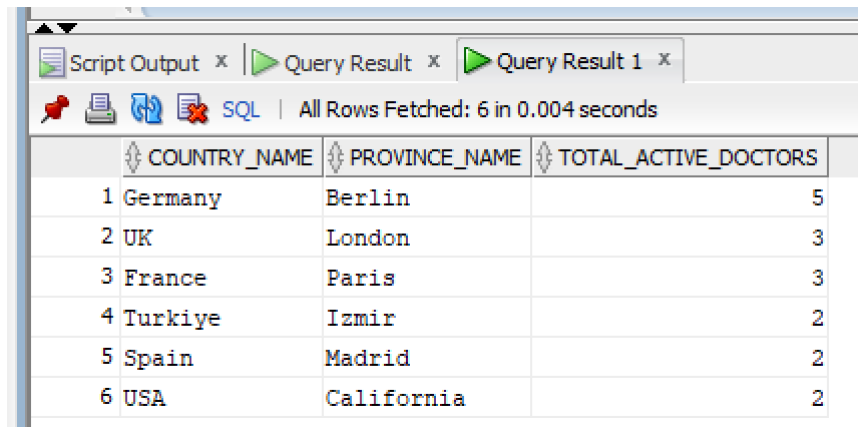
Problem

Display the number of active doctors in each country and province, showing only provinces with more than one active doctor.

Query

```
SELECT c.country_name,
       p.province_name,
       COUNT(d.doctor_id) AS total_active_doctors
FROM Country c
JOIN Province p ON c.country_id = p.country_id
JOIN District di ON p.province_id = di.province_id
JOIN Neighbourhood n ON di.district_id = n.district_id
JOIN Address a ON n.neighbourhood_id = a.neighbourhood_id
JOIN Doctor d ON a.address_id = d.address_id
JOIN Job_Status js ON d.job_status_id = js.job_status_id
WHERE js.status_name = 'ACTIVE'
GROUP BY c.country_name,
         p.province_name
HAVING COUNT(d.doctor_id) > 1
ORDER BY total_active_doctors DESC;
```

Results



	COUNTRY_NAME	PROVINCE_NAME	TOTAL_ACTIVE_DOCTORS
1	Germany	Berlin	5
2	UK	London	3
3	France	Paris	3
4	Turkiye	Izmir	2
5	Spain	Madrid	2
6	USA	California	2

Shehab

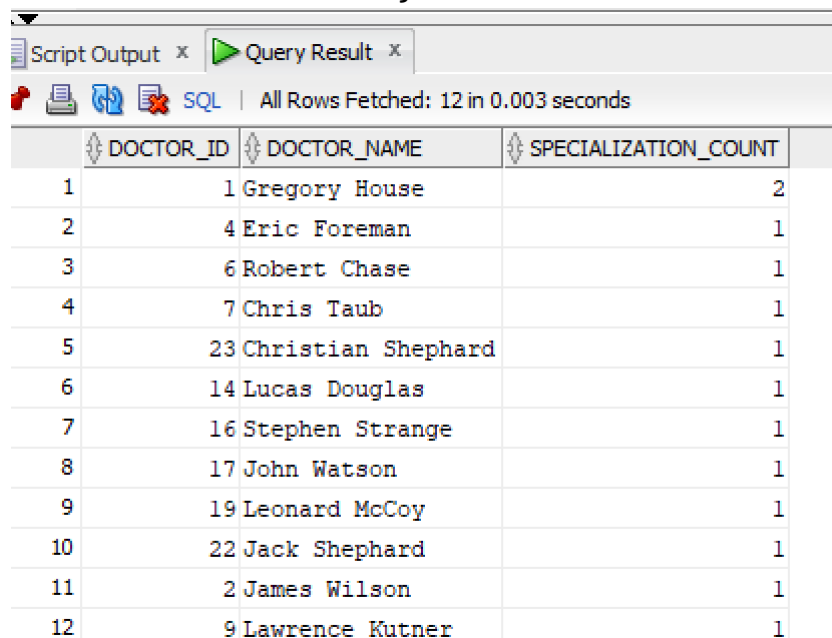
Problem

Display all male doctors who have at least one specialization, showing the number of specializations for each doctor, sorted by specialization count in descending order.

Query

```
SELECT d.doctor_id,  
       d.doc_first_name || ' ' || d.doc_last_name AS doctor_name,  
       COUNT(ds.spec_id) AS specialization_count  
FROM Doctor d  
JOIN Doctor_Specialization ds ON d.doctor_id = ds.doctor_id  
WHERE d.doc_gender = 'M'  
AND d.doctor_id IN (  
    SELECT doctor_id  
    FROM Doctor_Specialization  
    GROUP BY doctor_id  
    HAVING COUNT(spec_id) >=1  
)  
GROUP BY d.doctor_id, d.doc_first_name, d.doc_last_name  
HAVING COUNT(ds.spec_id) >= 1  
ORDER BY specialization_count DESC;
```

Results



	DOCTOR_ID	DOCTOR_NAME	SPECIALIZATION_COUNT
1	1	Gregory House	2
2	4	Eric Foreman	1
3	6	Robert Chase	1
4	7	Chris Taub	1
5	23	Christian Shephard	1
6	14	Lucas Douglas	1
7	16	Stephen Strange	1
8	17	John Watson	1
9	19	Leonard McCoy	1
10	22	Jack Shephard	1
11	2	James Wilson	1
12	9	Lawrence Kutner	1

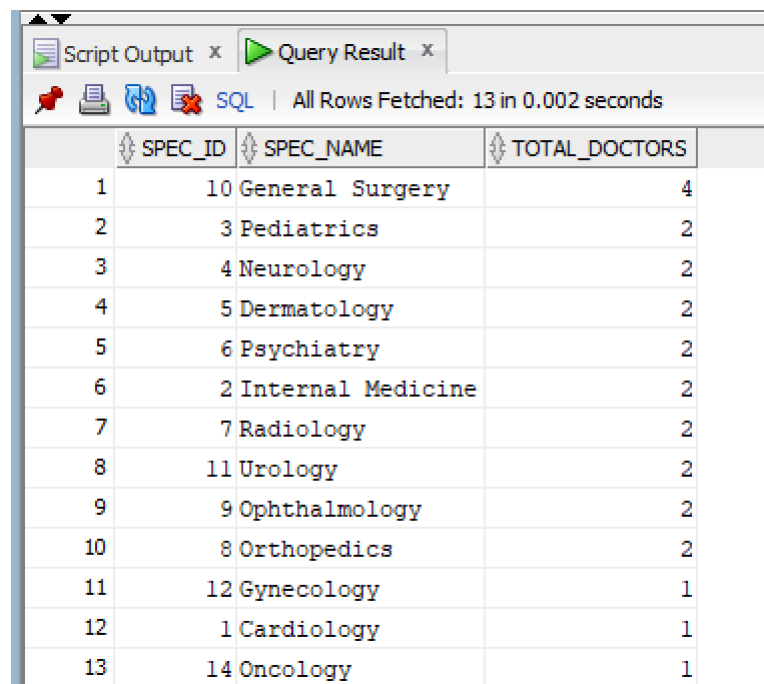
Problem

Display all specializations that are assigned to at least one doctor, showing the number of doctors in each specialization, sorted by the number of doctors in descending order.

Query

```
SELECT s.spec_id, s.spec_name,  
       COUNT(ds.doctor_id) AS total_doctors  
FROM Specialization s  
LEFT JOIN Doctor_Specialization ds ON s.spec_id = ds.spec_id  
WHERE s.spec_id IN (  
    SELECT spec_id  
    FROM Doctor_Specialization  
    GROUP BY spec_id  
    HAVING COUNT(doctor_id) >= 1  
)  
GROUP BY s.spec_id, s.spec_name  
HAVING COUNT(ds.doctor_id) >= 1  
ORDER BY total_doctors DESC;
```

Results



The screenshot shows a database query result window with two tabs: 'Script Output' and 'Query Result'. The 'Query Result' tab is active, displaying a table with 13 rows. The table has three columns: 'SPEC_ID', 'SPEC_NAME', and 'TOTAL_DOCTORS'. The data is sorted by 'TOTAL_DOCTORS' in descending order. The first row has 4 doctors, and the remaining 12 rows have 2 or 1 doctor each.

SPEC_ID	SPEC_NAME	TOTAL_DOCTORS
1	10 General Surgery	4
2	3 Pediatrics	2
3	4 Neurology	2
4	5 Dermatology	2
5	6 Psychiatry	2
6	2 Internal Medicine	2
7	7 Radiology	2
8	11 Urology	2
9	9 Ophthalmology	2
10	8 Orthopedics	2
11	12 Gynecology	1
12	1 Cardiology	1
13	14 Oncology	1

Hamza

Problem

Display all active doctors who have handled more than two active appointments. Show the doctor's ID, full name, department name, and the total number of appointments handled. Sort the results by the number of appointments in descending order and then by doctor name.

Query

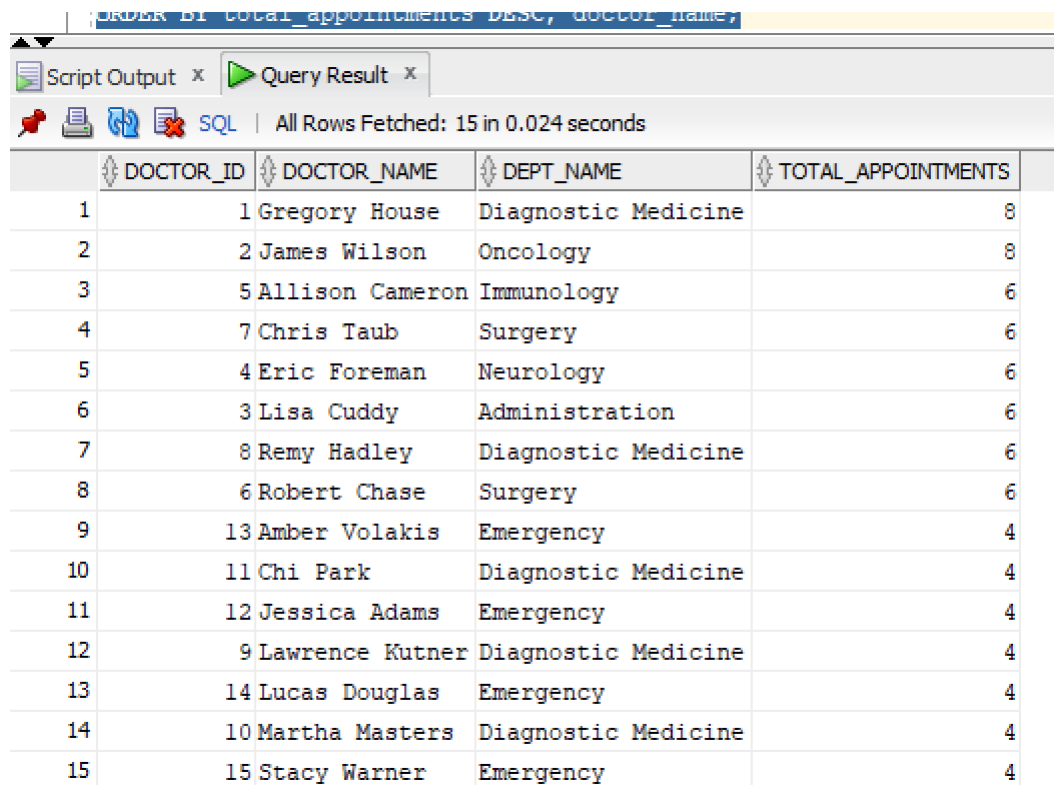
```
SELECT  
d.doctor_id, d.doc_first_name || ' ' || d.doc_last_name AS doctor_name,
```

```

    dep.dept_name, COUNT(a.appointment_id) AS total_appointments
FROM Appointment a
JOIN Doctor d ON a.doctor_id = d.doctor_id
JOIN Department dep ON d.dept_id = dep.dept_id
WHERE a.is_active = 1
GROUP BY d.doctor_id, d.doc_first_name, d.doc_last_name, dep.dept_name
HAVING COUNT(a.appointment_id) > 2
ORDER BY total_appointments DESC, doctor_name;

```

Results



	DOCTOR_ID	DOCTOR_NAME	DEPT_NAME	TOTAL_APPOINTMENTS
1	1	Gregory House	Diagnostic Medicine	8
2	2	James Wilson	Oncology	8
3	5	Allison Cameron	Immunology	6
4	7	Chris Taub	Surgery	6
5	4	Eric Foreman	Neurology	6
6	3	Lisa Cuddy	Administration	6
7	8	Remy Hadley	Diagnostic Medicine	6
8	6	Robert Chase	Surgery	6
9	13	Amber Volakis	Emergency	4
10	11	Chi Park	Diagnostic Medicine	4
11	12	Jessica Adams	Emergency	4
12	9	Lawrence Kutner	Diagnostic Medicine	4
13	14	Lucas Douglas	Emergency	4
14	10	Martha Masters	Diagnostic Medicine	4
15	15	Stacy Warner	Emergency	4

Problem

Show the total treatments, total revenue, and total tax generated by each department.

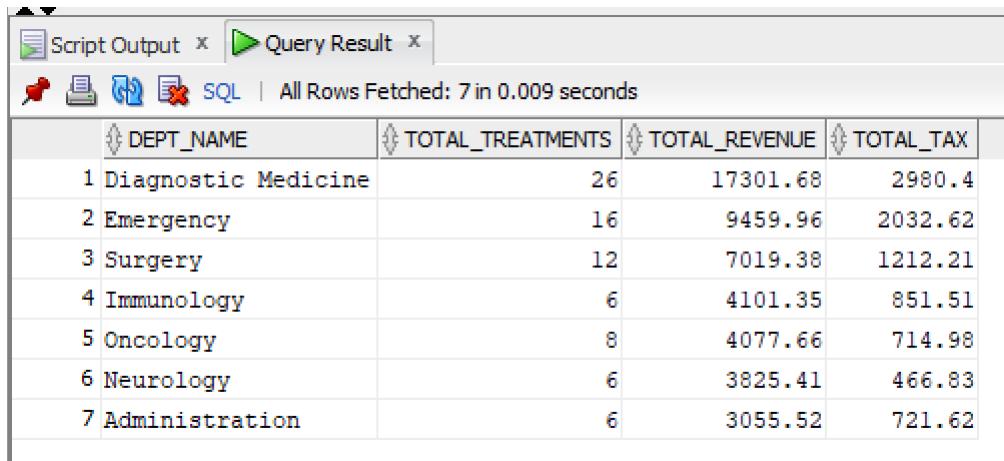
Query

```

SELECT
    dep.dept_name, COUNT(DISTINCT t.treatment_id) AS total_treatments,
    SUM(b.total_amount) AS total_revenue, SUM(b.tax) AS total_tax
FROM Department dep
JOIN Doctor d ON dep.dept_id = d.dept_id
JOIN Appointment a ON d.doctor_id = a.doctor_id
JOIN Treatment t ON a.appointment_id = t.appointment_id
JOIN Bill b ON t.treatment_id = b.treatment_id
WHERE a.is_active = 1
GROUP BY dep.dept_name
HAVING SUM(b.total_amount) > 0
ORDER BY total_revenue DESC;

```

Results



	DEPT_NAME	TOTAL_TREATMENTS	TOTAL_REVENUE	TOTAL_TAX
1	Diagnostic Medicine	26	17301.68	2980.4
2	Emergency	16	9459.96	2032.62
3	Surgery	12	7019.38	1212.21
4	Immunology	6	4101.35	851.51
5	Oncology	8	4077.66	714.98
6	Neurology	6	3825.41	466.83
7	Administration	6	3055.52	721.62

Bahar

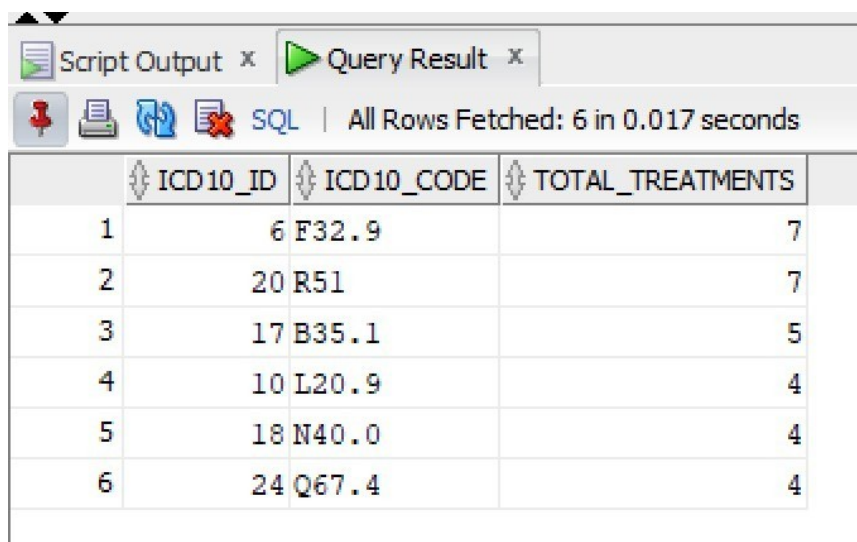
Problem

Display ICD10 codes with IDs greater than 5 that are associated with more treatments than the average number of treatments per ICD10 code.

Query

```
SELECT i.icd10_id, i.icd10_code, COUNT(t.treatment_id) AS
total_treatments
FROM ICD10 i
JOIN Treatment t ON i.icd10_id = t.icd10_id
WHERE i.icd10_id > 5
GROUP BY i.icd10_id, i.icd10_code
HAVING COUNT(t.treatment_id) > (
    SELECT AVG(cnt)
    FROM (
        SELECT COUNT(*) AS cnt
        FROM Treatment
        GROUP BY icd10_id
    )
)
ORDER BY total_treatments DESC;
```

Results



	ICD10_ID	ICD10_CODE	TOTAL_TREATMENTS
1	6	F32.9	7
2	20	R51	7
3	17	B35.1	5
4	10	L20.9	4
5	18	N40.0	4
6	24	Q67.4	4

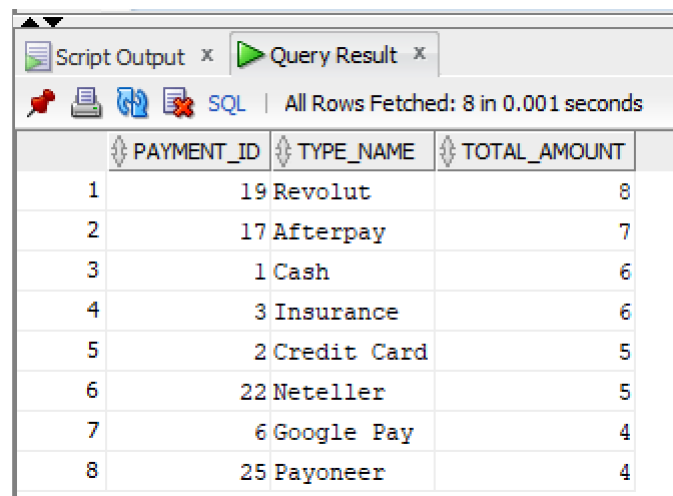
Problem

Display payment methods whose total bill count is above the average number of bills across all payment methods.

Query

```
SELECT pm.payment_id, pm.type_name, COUNT(b.treatment_id) AS
total_amount
FROM Payment_Method pm
JOIN Bill b ON pm.payment_id = b.Payment_id
GROUP BY pm.payment_id, pm.type_name
HAVING COUNT(b.treatment_id) > (
    SELECT AVG(bill_count)
    FROM (
        SELECT COUNT(*) AS bill_count
        FROM Bill
        GROUP BY payment_id
    )
)
ORDER BY total_amount DESC;
```

Results



The screenshot shows a database query result window with two tabs: 'Script Output' and 'Query Result'. The 'Query Result' tab is active, displaying a table with 8 rows. The table has three columns: 'PAYMENT_ID', 'TYPE_NAME', and 'TOTAL_AMOUNT'. The data is sorted by 'TOTAL_AMOUNT' in descending order. The status bar indicates 'All Rows Fetched: 8 in 0.001 seconds'.

	PAYMENT_ID	TYPE_NAME	TOTAL_AMOUNT
1	19	Revolut	8
2	17	Afterpay	7
3	1	Cash	6
4	3	Insurance	6
5	2	Credit Card	5
6	22	Neteller	5
7	6	Google Pay	4
8	25	Payoneer	4